

READING 28

PRICING AND VALUATION OF FORWARD COMMITMENTS

EXAM FOCUS

This topic review covers the calculation of price and value for forward and future contracts, specifically equity forward contracts, bond forward contracts, and forward (interest) rate agreements. This topic review also covers pricing and valuation of different swaps, including interest rate swaps, equity swaps, and currency swaps. Be able to structure profitable arbitrage transactions should the forward price differs from the no-arbitrage price. There are many formulas in this reading. However, once you realize the fundamental principles of no-arbitrage pricing, the formulas will start becoming somewhat less intimidating. There is a lot of testable material from this seven-LOS reading!

MODULE 28.1: PRICING AND VALUATION CONCEPTS



Video covering
this content is
available online.

Warm-Up: Forward Contracts

The party to the forward contract that agrees to buy the financial or physical asset has a **long forward position** and is called the *long*. The party to the forward contract that agrees to sell/deliver the asset has a **short forward position** and is called the *short*.

We will illustrate the basic forward contract mechanics through an example based on the purchase and sale of a Treasury bill. Note that while forward contracts on T-bills are usually quoted in terms of a discount percentage from face value, we use dollar prices here to make the example easy to follow.

Consider a contract under which Party A agrees to buy a \$1,000 face value 90-day Treasury bill from Party B 30 days from now at a price of \$990. Party A is the long and Party B is the short. Both parties have removed uncertainty about the price they will pay or receive for the T-bill at the future date. If 30 days from now T-bills are trading at \$992, the short must deliver the T-bill to the long in exchange for a \$990 payment. If T-bills are trading at \$988 on the future date, the long must purchase the T-bill from the short for \$990, the contract price.

Typically, no money changes hands at the inception of the contract, unlike futures contracts in which each party posts an initial deposit called the **margin** as a guarantee of performance.

At any point in time, including the settlement date, the party to the forward contract with the negative value will owe money to the other side. The other side of the contract will have a positive value of equal amount. Following this example, if the T-bill price is \$992 at the (future) settlement date, and the short does not deliver the T-bill for \$990 as promised, the short has defaulted.

Forward Contract Price Determination

The No-Arbitrage Principle

The price of a forward contract is *not* the price to purchase the contract; in fact, the parties to a forward contract typically pay nothing to enter into the contract at its inception. Here, *price refers to the forward price of the underlying*. This price may be a U.S. dollar or euro price, but it is often expressed as an interest rate or currency exchange rate. For T-bills, the price will be expressed as an annualized percentage discount from face value; for coupon bonds, it will usually be expressed as a yield to maturity; for the implicit loan in a forward rate agreement (FRA), it will be expressed as an annualized MRR; and for a currency forward, it is expressed as an exchange rate between the two currencies involved. However it is expressed, this rate, yield, discount, or dollar amount is the forward price in the contract.

The price that we wish to determine is the forward price that makes the *values* of both the long and the short positions zero at contract initiation. We will use the **no-arbitrage principle**: there should be no riskless profit to be gained by a combination of a forward contract position with positions in other assets. This principle assumes that (1) transactions costs are zero, (2) there are no restrictions on short sales or on the use of short sale proceeds, and (3) both borrowing and lending can be done in unlimited amounts at the risk-free rate of interest. This concept is so important that we'll express it in a formula:

forward price = price that prevents profitable riskless arbitrage in frictionless markets



PROFESSOR'S NOTE

We cover the first two LOS out of curriculum order for ease of exposition.

LOS 28.b: Describe the carry arbitrage model without underlying cashflows and with underlying cashflows.

A Simple Version of the Cost-of-Carry Model

In order to explain the no-arbitrage condition as it applies to the determination of forward prices, we will first consider a forward contract on an asset that costs nothing to store and makes no payments to its owner over the life of the forward contract. A zero-coupon (pure discount) bond meets these criteria. Unlike gold or wheat, it has no storage costs; unlike stocks, there are no dividend payments to consider; and unlike coupon bonds, it makes no periodic interest payments.

The general form for the calculation of the forward contract price can be stated as follows:

$$FP = S_0 \times (1 + R_f)^T$$

or

$$S_0 = \frac{FP}{(1 + R_f)^T}$$

where:

FP = forward price

S_0 = spot price at inception of the contract ($t = 0$)

R_f = annual risk-free rate

T = forward contract term in years

EXAMPLE: Calculating the no-arbitrage forward price

Consider a 3-month forward contract on a zero-coupon bond with a face value of \$1,000 that is currently quoted at \$500, and suppose that the annual risk-free rate is 6%. Determine the price of the forward contract under the no-arbitrage principle.

Answer:

$$T = \frac{3}{12} = 0.25$$

$$FP = S_0 \times (1 + R_f)^T = \$500 \times 1.06^{0.25} = \$507.34$$

Now, let's explore in more detail why \$507.34 is the no-arbitrage price of the forward contract.

Cash and Carry Arbitrage When the Forward Contract Is Overpriced

Suppose the forward contract is actually trading at \$510 rather than the no-arbitrage price of \$507.34. A short position in the forward contract requires the delivery of this bond three months from now. The arbitrage that we examine in this case amounts to borrowing \$500 at the risk-free rate of 6%, buying the bond for \$500, and simultaneously taking the short position in the forward contract on the zero-coupon bond so that we are obligated to deliver the bond at the expiration of the contract for the forward price of \$510.

At the settlement date, we can satisfy our obligation under the terms of the forward contract by delivering the zero-coupon bond for a payment of \$510, regardless of its market value at that time. We will use the \$510 payment we receive at settlement from the forward contract (the forward contract price) to repay the \$500 loan. The total amount to repay the loan, since the term of the loan is three months, is:

$$\text{loan repayment} = \$500 \times (1.06)^{0.25} = \$507.34$$

The payment of \$510 we receive when we deliver the bond at the forward price is greater than our loan payoff of \$507.34, and we will have earned an arbitrage profit of $\$510 - \$507.34 = \$2.66$. Notice that this is equal to the difference between the actual forward price and the no-arbitrage forward price. The transactions are illustrated in Figure 28.1.

Figure 28.1: Cash and Carry Arbitrage When Forward Is Overpriced

Today		Three Months From Today	
Spot price of bond	\$500		
Forward price	\$510		
<i>Transaction</i>	<i>Cash flow</i>	<i>Transaction</i>	<i>Cash flow</i>
Short forward	\$0	Settle short position by delivering bond	\$510.00
Buy bond	-\$500		
Borrow at 6%	+\$500	Repay loan	-\$507.34
Total cash flow	\$0	Total cash flow = arbitrage profit	+\$2.66



PROFESSOR'S NOTE

Here's a couple hints to help you remember which transactions to undertake for cash and carry arbitrage: (1) always start with nothing (no cash, no securities), (2) buy underpriced assets and sell overpriced assets ("buy low, sell high"), and (3) take opposite positions in the spot and forward markets.

So, if the futures contract is overpriced, you want to take a short position in those futures (which obligates you to sell at a fixed price). Because you go short in the forward market, you take the opposite position in the spot market and buy the asset. You need money to buy the asset, so you have to borrow. Therefore, to set up a cash and carry arbitrage:

forward overpriced:

borrow money $\Rightarrow$ buy (go long) the spot asset $\Rightarrow$ go short the asset in the forward market

So far, the underlying instrument was a zero-coupon bond, and hence had no associated cash flows during the forward contract period. If the holder of the underlying receives cash flows (as coupon or dividends), the present value of those cash flows will be subtracted from the spot price of the underlying. We will see this later in this reading.

Reverse Cash and Carry Arbitrage When the Forward Contract Is Underpriced

Suppose the forward contract is actually trading at \$502 instead of the no-arbitrage price of \$507.34. We reverse the arbitrage trades from the previous case and generate an arbitrage profit as follows: We sell the bond short today for \$500 and simultaneously take the long position in the forward contract, which obligates us to purchase the bond in 90 days at the forward price of \$502. We invest the \$500 proceeds from the short sale at the 6% annual rate for three months.

In this case, at the settlement date, we receive the investment proceeds of \$507.34, accept delivery of the bond in return for a payment of \$502, and close out our short position by delivering the bond we just purchased at the forward price.

The payment of \$502 we make as the long position in the contract is less than the investment proceeds of \$507.34, and we have earned an arbitrage profit of $\$507.34 - \$502 = \$5.34$. The transactions are illustrated in Figure 28.2.

Figure 28.2: Reverse Cash and Carry Arbitrage When Forward Is Underpriced

Today		Three Months From Today	
Spot price of bond		\$500	
Forward price		\$502	
<i>Transaction</i>	<i>Cash flow</i>	<i>Transaction</i>	<i>Cash flow</i>
Long forward	\$0	Settle long position by buying bond	-\$502.00
Short sell bond	+\$500	Deliver bond to close short position	\$0.00
Invest short-sale proceeds at 6%	-\$500	Receive investment proceeds	+\$507.34
Total cash flow	\$0	Total cash flow = arbitrage profit	+\$5.34

In this case, because the forward contract is underpriced, the trades are reversed from cash and carry arbitrage. To set up the reverse cash-and-carry arbitrage:

forward underpriced:

borrow asset $\Rightarrow$ short (sell) spot asset $\Rightarrow$ lend money $\Rightarrow$ long (buy) forward

From the calculations just listed, we can see that the no-arbitrage forward price that yields a zero *value* for both the long and short positions in the forward contract at inception is the no-arbitrage price of \$507.34.



PROFESSOR'S NOTE

Day count and compounding conventions vary among different financial instruments. There are three variations used in the CFA curriculum:

- All MRR-based contracts such as FRAs, swaps, caps, floors, etc.:
 - 360 days per year and simple interest
 - Multiply “r” by days/360
- Equities, bonds, and stock options:
 - 365 days per year and periodic compound interest
 - Raise $(1 + r)$ to an exponent of days/365
- Equity indexes:
 - 365 days per year and continuous compounding
 - Raise Euler’s number “e” to an exponent of “r” times days/365

MODULE 28.2: PRICING AND VALUATION OF EQUITY FORWARDS



Video covering this content is available online.

LOS 28.a: Describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value.

Equity Forward Contracts With Discrete Dividends

Recall that the no-arbitrage forward price in our earlier example was calculated for an asset with no periodic payments. A stock, a stock portfolio, or an equity index may have expected

dividend payments over the life of the contract. In order to price such a contract, we must either adjust the spot price for the present value of the expected dividends (PVD) over the life of the contract, or adjust the forward price for the future value of the dividends (FVD) over the life of the contract. The **no-arbitrage price of an equity forward contract** in either case is:

$$FP \text{ (of an equity security)} = (S_0 - PVD) \times (1 + R_f)^T$$

$$FP \text{ (of an equity security)} = [S_0 \times (1 + R_f)^T] - FVD$$

For equity contracts, use a 365-day basis for calculating T . For example, if it is a 60-day contract, $T = 60 / 365$.

EXAMPLE: Calculating the price of a forward contract on a stock

Calculate the no-arbitrage forward price for a 100-day forward on a stock that is currently priced at \$30.00 and is expected to pay a dividend of \$0.40 in 15 days, \$0.40 in 85 days, and \$0.50 in 175 days. The annual risk-free rate is 5%, and the yield curve is flat.

Answer:

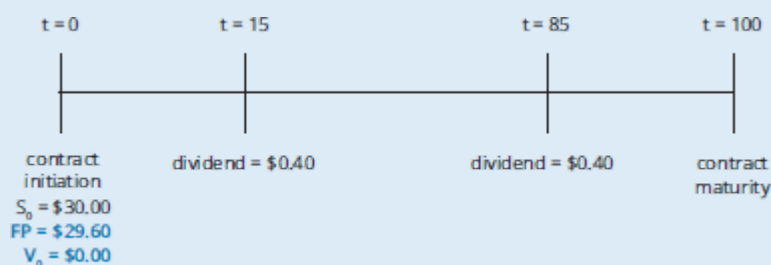
Ignore the dividend in 175 days because it occurs after the maturity of the forward contract.

$$PVD = \frac{\$0.40}{1.05^{15/365}} + \frac{\$0.40}{1.05^{85/365}} = \$0.7946$$

$$FP = (\$30.00 - \$0.7946) \times 1.05^{100/365} = \$29.60$$

The timeline of cash flows is shown in the following figure.

Pricing a 100-Day Forward Contract on Dividend-Paying Stock



To calculate the value of the long position in a **forward contract on a dividend-paying stock**, we make the adjustment for the present value of the remaining expected discrete dividends at time t (PVD_t) to get:

$$V_t(\text{long position}) = [S_t - PVD_t] - \left[\frac{FP}{(1 + R_f)^{(T-t)}} \right]$$



PROFESSOR'S NOTE

This formula still looks like the standard spot price minus present value of forward price. However, now the “spot price” has been adjusted by subtracting out the present value of the dividends because the long position in the forward

contract does not receive the dividends paid on the underlying stock. So, now think adjusted spot price less present value of forward price.

If given the current forward price (FP_t) on the same underlying and with the same maturity:

$$V_t(\text{long position}) = \left[\frac{FP_t - FP}{(1 + R_f)^{T-t}} \right]$$

EXAMPLE: Calculating the value of an equity forward contract on a stock

After 60 days, the value of the stock in the previous example is \$36.00. Calculate the value of the equity forward contract on the stock to the long position, assuming the risk-free rate is still 5% and the yield curve is flat.

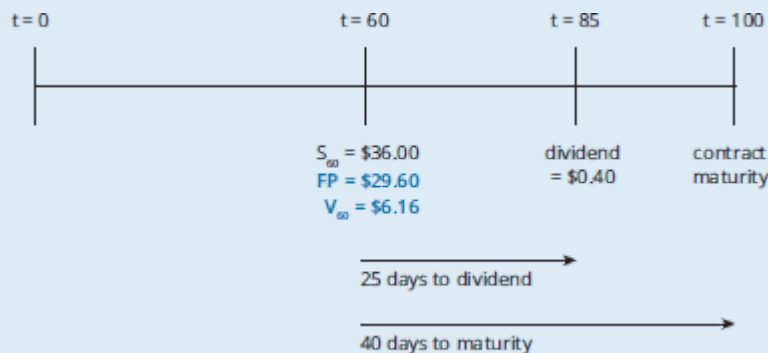
Answer:

There's only one dividend remaining (in 25 days) before the contract matures (in 40 days) as shown here, so:

$$PVD_{60} = \frac{\$0.40}{1.05^{25/365}} = \$0.3987$$

$$V_{60}(\text{long position}) = \$36.00 - \$0.3987 - \left[\frac{\$29.60}{1.05^{40/365}} \right] = \$6.16$$

Valuing a 100-Day Forward Contract After 60 Days



Equity Forward Contracts With Continuous Dividends

To calculate the price of an **equity index forward contract**, rather than take the present value of each dividend on (possibly) hundreds of stocks, we can make the calculation as if the dividends are paid continuously (rather than at discrete times) at the dividend yield rate on the index. Using continuous time discounting, we can calculate the no-arbitrage forward price as:

$$FP(\text{on an equity index}) = S_0 \times e^{(R_f^c - \delta^c) \times T} = (S_0 \times e^{\delta^c \times T}) \times e^{R_f^c \times T}$$

where:

R_f^c = continuously compounded risk-free rate

δ^c = continuously compounded dividend yield



PROFESSOR'S NOTE

The relationship between the periodically compounded risk-free rate R_f and the continuously compounded rate R_f^c is $R_f^c = R_f \times \ln(1 + R_f)$. For example, 5% compounded annually is equal to $\ln(1.05) = 0.04879 = 4.879\%$ compounded continuously. The 2-year 5% future value factor can then be calculated as either $1.05^2 = 1.1025$ or $e^{0.04879 \times 2} = 1.1025$.

EXAMPLE: Calculating the price of a forward contract on an equity index

The value of the S&P 500 Index is 1,140. The continuously compounded risk-free rate is 4.6% and the continuous dividend yield is 2.1%. Calculate the no-arbitrage price of a 140-day forward contract on the index.

Answer:

$$FP = 1,140 \times e^{(0.046 - 0.021) \times (140/365)} = 1,151$$

On a TI BA II PLUS calculator, use the following keystrokes:

0.046[-]0.021[=][×]140[÷]365[=][2nd][LN][×]1140[=]



MODULE QUIZ 28.1, 28.2

1. A stock is currently priced at \$30 and is expected to pay a dividend of \$0.30 20 days and 65 days from now. The contract price for a 60-day forward contract when the interest rate is 5% is *closest* to:
 - A. \$29.46.
 - B. \$29.70.
 - C. \$29.94.
2. After 37 days, the stock in Question 1 is priced at \$21, and the risk-free rate is still 5%. The value of the forward contract on the stock to the short position is:
 - A. -\$8.85.
 - B. +\$8.85.
 - C. +\$9.00.
3. The forward price of a 200-day stock index futures contract when the spot index is 540, the continuous dividend yield is 1.8%, and the continuously compounded risk-free rate is 7% (with a flat yield curve) is *closest* to:
 - A. 545.72.
 - B. 555.61.
 - C. 568.08.
4. An analyst who mistakenly ignores the dividends when valuing a short position in a forward contract on a stock that pays dividends will *most likely*:
 - A. overvalue the position by the present value of the dividends.
 - B. undervalue the position by the present value of the dividends.
 - C. overvalue the position by the future value of the dividends.
5. A portfolio manager owns Macrogrow, Inc., which is currently trading at \$35 per share. She plans to sell the stock in 120 days, but is concerned about a possible price decline. She decides to take a short position in a 120-day forward contract on the stock. The stock will pay a \$0.50 per share dividend in 35 days and \$0.50 again in 125 days. The risk-free rate is 4%. The value of the trader's position in the forward contract in 45 days, assuming in 45 days the stock price is \$27.50 and the risk-free rate has not changed, is *closest* to:

- A. \$7.17.
- B. \$7.50.
- C. \$7.92.

MODULE 28.3: PRICING AND VALUATION OF FIXED INCOME FORWARDS



Video covering
this content is
available online.

LOS 28.d: Describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value.

Forwards and Futures on Fixed Income Securities

In order to calculate the no-arbitrage **forward price on a coupon-paying bond**, we can use the same formula as we used for a dividend-paying stock or portfolio, simply substituting the present value of the expected coupon payments (PVC) *over the life of the contract* for PVD, or the future value of the coupon payments (FVC) for FVD, to get the following formulas:

$$\begin{aligned} \text{FP(on a fixed income security)} &= (S_0 - \text{PVC}) \times (1 + R_f)^T \\ \text{or} \\ &= S_0 \times (1 + R_f)^T - \text{FVC} \end{aligned}$$

The value of the forward contract prior to expiration is as follows:

$$V_t(\text{long position}) = [S_t - \text{PVC}_t] - \left[\frac{\text{FP}}{(1 + R_f)^{(T-t)}} \right]$$

If given the current forward price (FP_t) on the same underlying and with the same maturity:

$$V_t(\text{long position}) = \left[\frac{\text{FP}_t - \text{FP}}{(1 + R_f)^{(T-t)}} \right]$$

In our examples, we assume that the spot price of the underlying coupon-paying bond includes accrued interest. For fixed income contracts, use a 365-day basis to calculate T if the contract maturity is given in days.

EXAMPLE: Calculating the price of a forward on a fixed income security

Calculate the price of a 250-day forward contract on a 7% U.S. Treasury bond with a spot price of \$1,050 (including accrued interest) that has just paid a coupon and will make another coupon payment in 182 days. The annual risk-free rate is 6%.

Answer:

Remember that U.S. Treasury bonds make semiannual coupon payments, so:

$$C = \frac{\$1,000 \times 0.07}{2} = \$35.00$$

$$PVC = \frac{\$35.00}{1.06^{182/365}} = \$34.00$$

The forward price of the contract is therefore:

$$FP \text{ (on a fixed income security)} = (\$1,050 - \$34.00) \times 1.06^{250/365} = \$1,057.37$$

EXAMPLE: Calculating the value of a forward on a fixed income security

After 100 days, the value of the bond in the previous example is \$1,090. Calculate the value of the forward contract on the bond to the long position, assuming the risk-free rate is 6.0%.

Answer:

There is only one coupon remaining (in 82 days) before the contract matures (in 150 days), so:

$$PVC = \frac{\$35.00}{1.06^{82/365}} = \$34.54$$

$$V_{100} \text{ (long position)} = \$1,090 - \$34.54 - \left(\frac{\$1,057.37}{1.06^{150/365}} \right) = \$23.11$$

If given the forward price directly, the value (to the long party) is simply the present value of the difference between the current forward price and the original, locked-in forward price.

EXAMPLE: Value of a fixed-income forward contract after inception

Louise Michelle entered into a 250-day forward contract on a 7% U.S. Treasury bond at a forward price of \$1057.37. 100 days later, the forward price has changed to \$1081.04. The risk-free rate is 6%.

Calculate the value to Michelle of the forward contract.

Answer:

$$\text{value} = [1081.04 - 1057.37] / (1.06)^{(150/365)} = \$23.11$$

Bond futures contracts often allow the short an option to deliver any of several bonds, which will satisfy the delivery terms of the contract. This is called a delivery option and is valuable to the short. Each bond is given a conversion factor that is used to adjust the long's payment at delivery so the more valuable bonds receive a larger payment. These factors are multipliers for the futures price at settlement. The long pays the futures price at expiration, multiplied by the **conversion factor (CF)**.

Bond prices in some countries are quoted as **clean prices**. At settlement, the buyer actually pays the clean price plus an **accrued interest**, or the **full price**.

$$\text{accrued interest} = \left(\frac{\text{days since last coupon payment}}{\text{days between coupon payments}} \right) \times \text{coupon amount}$$

$$\text{full price} = \text{clean price} + \text{accrued interest} = \text{clean price} + AI_0$$

The futures price can then be calculated as:

$$FP = \left[(\text{full price})(1 + R_f)^T - FVC - AI_T \right]$$

where:

AI_T = the accrued interest at maturity of the futures contract

The quoted futures price adjusts this price based on the conversion factor (CF) as follows:

$$QFP = FP / CF = \left[(\text{full price})(1 + R_f)^T - FVC - AI_T \right] \left(\frac{1}{CF} \right)$$

EXAMPLE: Calculating the price of a Treasury bond futures contract

Suppose that you need to calculate the quoted futures price of a 1.2-year Treasury bond futures contract. The cheapest-to-deliver bond is a 7% T-bond with exactly 10 years remaining to maturity and a quoted price of \$1,040 with a conversion factor of 1.13. There is currently no accrued interest because the bond has just paid a coupon. The annual risk-free rate is 5%. The accrued interest on the bond at maturity of the futures contract will be \$14.

Answer:

The full price of the bond = \$1,040 quoted price + \$0 accrued interest = \$1,040. The semiannual coupon on a single, \$1,000 face-value 7% bond is \$35. A bondholder will receive one payment 0.5 years from now (when there are 0.7 years left to maturity of the futures contract) and one payment 1 year from now (when there are 0.2 years until maturity). The future value of these coupons at the end of 1.2 years (the expiration date) is:

$$FVC = (\$35 \times 1.05^{0.7}) + (\$35 \times 1.05^{0.2}) = \$71.56$$

The quoted futures price is then:

$$QFP = [(\$1,040 \times 1.05^{1.2}) - \$71.56 - \$14] \left(\frac{1}{1.13} \right) = \$900.13$$



MODULE QUIZ 28.3

1. A 6% Treasury bond is trading at \$1,044 (including accrued interest) per \$1,000 of face value. It will make a coupon payment 98 days from now. The yield curve is flat at 5% over the next 150 days. The forward price per \$1,000 of face value for a 120-day forward contract, is *closest* to:
 - A. \$1,014.52.
 - B. \$1,030.79.

MODULE 28.4: PRICING FORWARD RATE AGREEMENTS



Video covering
this content is
available online.

LOS 28.c: Describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value.

Warm-Up: MRR-Based Loans and Forward Rate Agreements

Eurodollar deposit is the term for deposits in large banks outside the United States denominated in U.S. dollars. The lending rate on dollar-denominated loans between banks is based on the **secured overnight funding rate (SOFR)**, hereafter referred to generically as the MRR (“market reference rate”). It is quoted as an annualized rate based on a 360-day year. In contrast to T-bill discount yields, MRR is an add-on rate, like a yield quote on a short-term certificate of deposit.

EXAMPLE: MRR-based loans

Compute the amount that must be repaid on a \$1 million loan for 30 days if 30-day MRR is quoted at 6%.

Answer:

The add-on interest is calculated as $\$1 \text{ million} \times 0.06 \times (30 / 360) = \$5,000$. The borrower would repay $\$1,000,000 + \$5,000 = \$1,005,000$ at the end of 30 days.

The long position in a **forward rate agreement (FRA)** is the party that is effectively borrowing money (long the loan, with the contract price being the interest rate on the loan). If the floating rate at contract expiration is above the rate specified in the forward agreement, the long position in the contract can be viewed as the right to borrow at below-market rates, and the long will receive a payment. If the floating rate at the expiration date is below the rate specified in the forward agreement, the short will receive a cash payment from the long. (The right to lend at *above*-market rates has a positive value.)



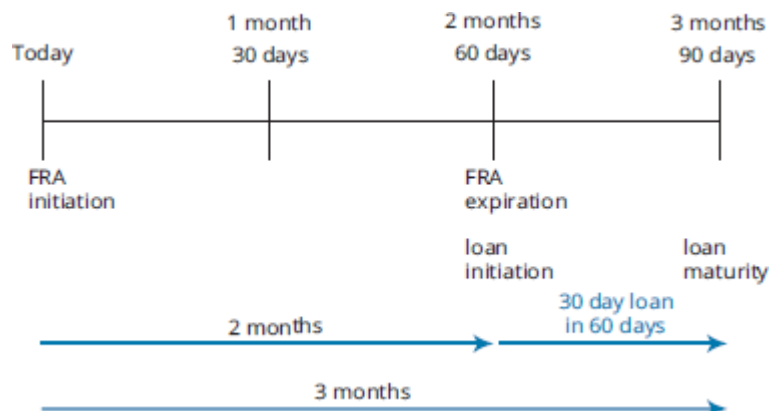
PROFESSOR'S NOTE

We say “can be viewed as” because an FRA is settled in cash, so there is no requirement to lend or borrow the amount stated in the contract. For this reason, the creditworthiness of the long position is not a factor in the determination of the interest rate on an FRA. However, to understand the pricing and calculation of value for an FRA, viewing the contract as a commitment to lend or borrow at a certain interest rate at a future date is helpful.

The notation for FRAs is unique. There are two numbers associated with an FRA: the number of months until the contract expires, and the number of months until the underlying loan is settled. The difference between these two is the maturity of the

underlying loan. For example, a 2×3 FRA is a contract that expires in two months (60 days), and the underlying loan is settled in three months (90 days). The underlying rate is 1-month (30-day) MRR on a 30-day loan in 60 days. See Figure 28.3.

Figure 28.3: Illustration of a 2×3 FRA



Pricing FRAs

There are three important things to remember about FRAs when we're pricing and valuing them:

1. MRR rates in the Eurodollar market are add-on rates and are always quoted on a 30/360 day basis in annual terms. For example, if the MRR quote on a 30-day loan is 6%, the actual unannualized monthly rate is $6\% \times (30/360) = 0.5\%$.
2. The long position in an FRA is, in effect, long the rate and benefits when the rate increases.
3. Although the interest on the underlying loan won't be paid until the end of the loan (e.g., in three months in Figure 28.3, the payoff on the FRA occurs at the expiration of the FRA (e.g., in two months). Therefore, the payoff on the FRA is the present value of the interest savings on the loan (e.g., discounted one month in Figure 28.3).

The forward "price" in an FRA is actually a forward interest rate. The calculation of a forward interest rate is presented in Level I as the computation of forward rates from spot rates. We will illustrate this calculation with an example.

EXAMPLE: Calculating the price of an FRA

Calculate the price of a 1×4 FRA (i.e., a 90-day loan, 30 days from now). The current 30-day MRR is 4% and 120-day MRR is 5%.

Answer:

The actual (unannualized) rate on the 30-day loan is:

$$R_{30} = 4\% \times \frac{30}{360} = 0.333\%$$

The actual (unannualized) rate on the 120-day loan is:

$$R_{120} = 5\% \times \frac{120}{360} = 1.667\%$$

We wish to calculate the actual rate on a 90-day loan from day 30 to day 120:

$$\text{price of } 1 \times 4 \text{ FRA} = \frac{1 + R_{120}}{1 + R_{30}} - 1 = \frac{1.01667}{1.00333} - 1 = 1.33\%$$

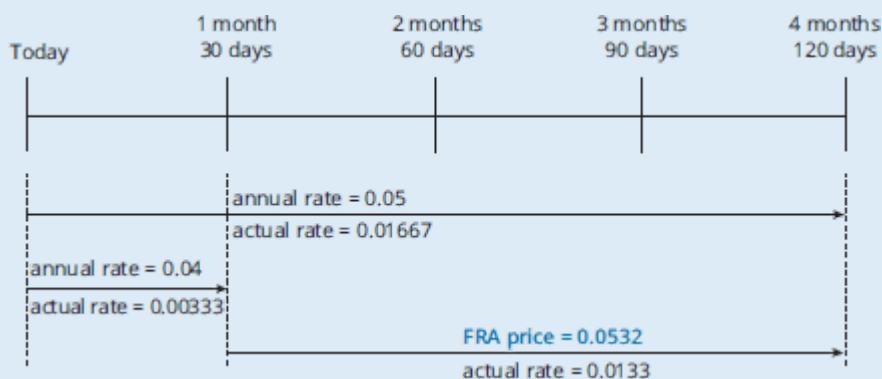
We can annualize this rate as:

$$1.33\% \times \frac{360}{90} = 0.0532 = 5.32\%$$

This is the no-arbitrage forward rate—the forward rate that will make the values of the long and the short positions in the FRA both zero at the initiation of the contract.

The time line is shown in the following figure.

Pricing a 1 × 4 FRA



MODULE QUIZ 28.4

- The contract rate (annualized) for a 3 × 5 FRA if the current 60-day rate is 4%, the current 90-day rate is 5%, and the current 150-day rate is 6%, is *closest* to:
 - 6.0%.
 - 6.9%.
 - 7.4%.
- The CFO of Yellow River Company wishes to hedge borrowing costs on a 90-day loan that starts in 90 days. The current term structure indicates the following: 30-day MRR is 4.5%, 90-day MRR is 4.7%, and 180-day MRR is 4.9%.

The appropriate hedging position to take, and the corresponding fair market rate, would *best* be described as:

- long 3 × 3 FRA at a rate of 4.48%.
- long 3 × 6 FRA at a rate of 4.48%.
- long 3 × 6 FRA at a rate of 5.02%.



Video covering
this content is
available online.

MODULE 28.5: VALUATION OF FORWARD RATE AGREEMENTS

Valuing an FRA at Maturity

To understand the calculation of the value of the FRA *after the initiation of the contract*, recall that in the previous example the long in the FRA has the “right” to borrow money 30 days from inception for a period of 90 days at the forward rate. If interest rates increase (specifically the 90-day forward contract rate), the long will profit as the contract has fixed a borrowing rate below the now-current market rate. These “savings” will come at the end of the loan term, so to value the FRA we need to take the present value of these savings. An example incorporating this fact will illustrate the cash settlement value of an FRA at expiration.

EXAMPLE: Calculating the value of an FRA at maturity (i.e., cash payment at settlement)

Continuing the prior example for a 1 × 4 FRA, assume a notional principal of \$1 million. Suppose that, at contract expiration, the 90-day rate has increased to 6%, which is above the contract rate of 5.32%. Calculate the value of the FRA at maturity, which is equal to the cash payment at settlement.

Answer:

The interest savings at the end of the loan term (compared to the market rate of 6%) will be:

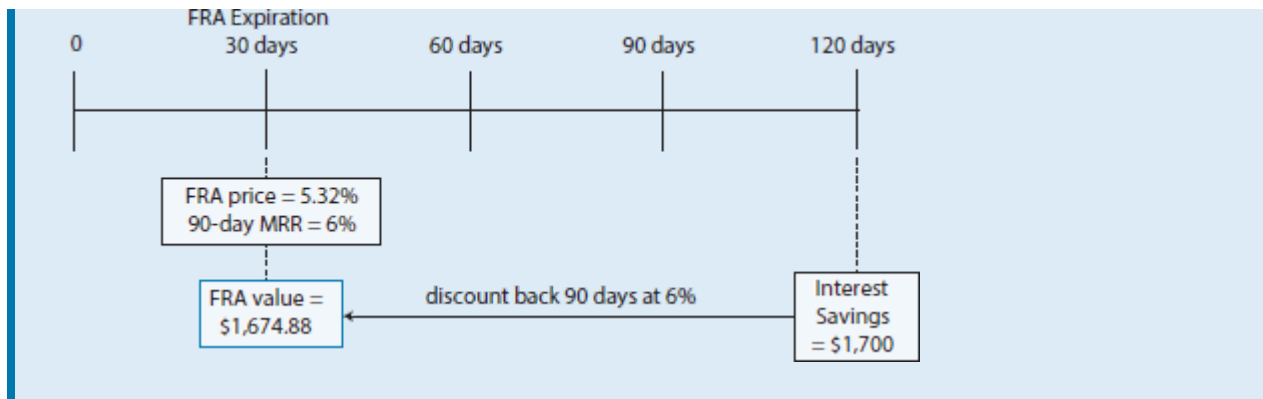
$$\left[\left(0.0600 \times \frac{90}{360} \right) - \left(0.0532 \times \frac{90}{360} \right) \right] \times \$1,000,000 = \$1,700$$

The present value of this amount at the FRA settlement date (90 days prior to the end of the loan term) discounted at the current rate of 6% is:

$$\frac{\$1,700}{1 + \left(0.06 \times \frac{90}{360} \right)} = \$1,674.88$$

This will be the cash settlement payment from the short to the long at the expiration of the contract. Note that we have discounted the savings in interest at the end of the loan term by the *market* rate of 6% that prevails at the contract settlement date for a 90-day term, as shown in the following figure.

Valuing a 1 × 4 FRA at Maturity



Valuing an FRA Before Maturity

To value an FRA *before* the settlement date, we need to know the number of days that have passed since the initiation of the contract. For example, let's suppose we want to value the same 1 × 4 FRA 10 days after initiation. Originally, it was a 1 × 4 FRA, which means the FRA now expires in 20 days. The calculation of the “savings” on the loan will be the same as in our previous example, except that we need to use the “new” FRA price that would be quoted on a contract covering the same period as the original “loan.” In this case the “new” FRA price is the now-current market forward rate for a 90-day loan made at the settlement date (20 days in the future). Also, we need to discount the interest savings implicit in the FRA back 110 days (i.e., an extra 20 days) instead of 90 days as we did for the value at the settlement date.

EXAMPLE: Calculating value of an FRA before settlement

Value a 5.32% 1 × 4 FRA with a principal amount of \$1 million 10 days after initiation if 110-day MRR is 5.9% and 20-day MRR is 5.7%.

Answer:

Step 1: Find the “new” FRA price on a 90-day loan 20 days from today. This is the current 90-day forward rate at the settlement date, 20 days from now.

$$\left[\frac{1 + \left(0.059 \times \frac{110}{360}\right)}{1 + \left(0.057 \times \frac{20}{360}\right)} - 1 \right] \times \frac{360}{90} = 0.0592568$$

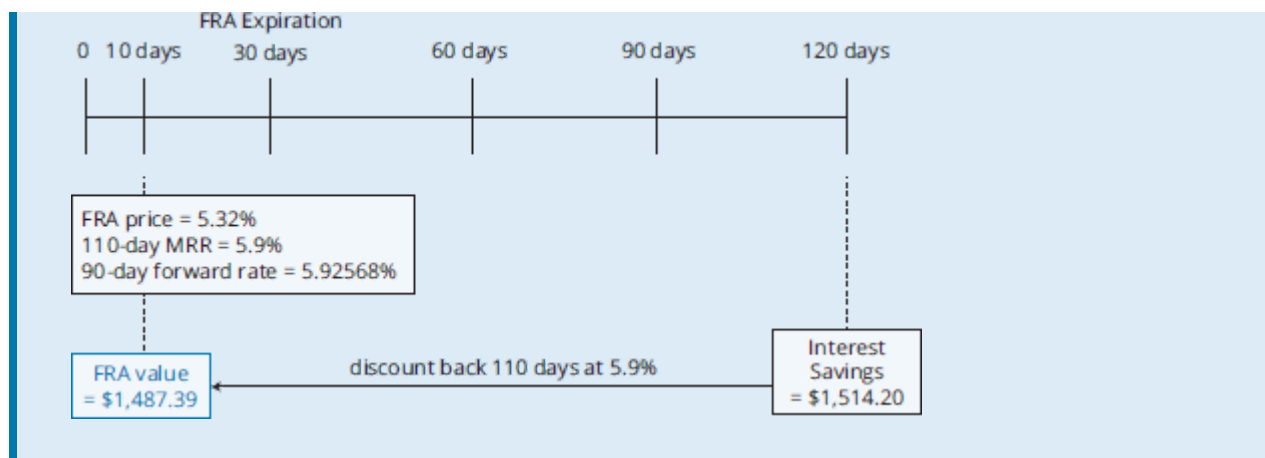
Step 2: Calculate the interest difference on a \$1 million, 90-day loan made 20 days from now at the forward rate calculated previously compared to the FRA rate of 5.32%.

$$\left[\left(0.0592568 \times \frac{90}{360}\right) - \left(0.0532 \times \frac{90}{360}\right) \right] \times \$1,000,000 = \$1,514.20$$

Step 3: Discount this amount at the current 110-day rate.

$$\frac{\$1,514.20}{1 + \left(0.059 \times \frac{110}{360}\right)} = \$1,487.39$$

Valuing a 1 × 4 FRA Before Settlement



PROFESSOR'S NOTE

I have tried to explain these calculations in such a way that you can value an FRA at any date from initiation to settlement using basic tools that you already know. Once you understand where the value of an FRA comes from (the interest savings on a loan to be made at the settlement date) and when this value is to be received (at the end of the loan), you can calculate the present value of these savings even under somewhat stressful test conditions. Just remember that if the rate in the future is less than the FRA rate, the long is "obligated to borrow" at above-market rates and will have to make a payment to the short. If the rate is greater than the FRA rate, the long will receive a payment from the short.

Futures Contracts

Futures contracts are very much like the forward contracts except that they trade on organized exchanges. Each exchange has a clearinghouse. The clearinghouse guarantees that traders in the futures market will honor their obligations. The clearinghouse does this by splitting each trade once it is made and acting as the opposite side of each position. To safeguard the clearinghouse, the exchange requires both sides of the trade to post margin and settle their accounts on a daily basis. Thus, the margin in the futures markets is a performance guarantee.

Marking to market is the process of adjusting the margin balance in a futures account each day for the change in the value of the contract from the previous trading day, based on the settlement price.

The pricing relationship discussed earlier also generally applies to futures contracts as well (see for example, our earlier discussion on Treasury bond futures contract). Like forward contracts, futures contracts have no value at contract initiation. Unlike forward contracts, futures contracts do not accumulate value changes over the term of the contract. Since futures accounts are marked to market daily, the value after the margin deposit has been adjusted for the day's gains and losses in contract value is always zero. The futures price at any point in time is the price that makes the value of a new contract equal to zero. The value of a futures contract strays from zero only during the trading periods between the times at which the account is marked to market:

$$\text{value of futures contract} = \text{current futures price} - \text{previous mark-to-market price}$$

If the futures price increases, the value of the long position increases. The value is set back to zero by the mark-to-market at the end of the mark-to-market period.



MODULE QUIZ 28.5

1. Ina bank entered into a 3×6 FRA at a rate of 3% on a \$10 million notional 30 days ago. Current MRR term structure is as follows:

Days	Rate
30	2.5%
60	2.6%
90	2.7%
120	2.9%
150	3.2%

The value of the long position in the FRA is *closest* to:

- A. 14,407.
- B. 14,612.
- C. 57,678.

MODULE 28.6: PRICING AND VALUATION OF INTEREST RATE SWAPS



Video covering this content is available online.

LOS 28.e: Describe how interest rate swaps are priced, and calculate and interpret their no-arbitrage value.

The distinction between the price and the value of a swap is the same one we made for forward contracts. Remember that the price of a forward contract is the forward rate that yields a zero value for the contract at initiation. After contract initiation, as rates or prices change, the contract will likely have value to either the long or the short.

Consider an interest rate swap. One party agrees to pay floating (borrow at the floating rate) and receive fixed (lend at a fixed rate). At initiation of the swap, the fixed rate is selected so that the present value of the floating-rate payments is equal to the present value of the fixed-rate payments, which means the swap value is zero to both parties. This fixed rate is called the **swap rate** or the **swap fixed rate**. Determining the swap rate is equivalent to “pricing” the swap.

As short-term rates (and expected future short-term rates) change over the term of the swap, the value of the swap to one of the parties will be positive. The swap position is an asset to that party.

If market interest rates increase during the life of the swap, the swap will take on a positive value from the perspective of the fixed-rate payer. The value of the swap contract to the other party (the pay-floating side) will thus be negative.

Computing the Swap Fixed Rate

The **swap fixed rate** is derived from the MRR curve corresponding to the swap tenor. Consider a two-year, semiannual interest rate swap. The swap fixed rate underlying this swap will be determined based on the MRR rates corresponding to the four settlement dates of this swap.

Using those four MRRs, we calculate the discount factors (Zs) for each.

$$Z = \frac{1}{1 + \left(\text{MRR} \times \frac{\text{days}}{360} \right)}$$

The periodic swap fixed rate SFR(periodic) can then be calculated as:

$$\begin{aligned} \text{SFR}(\text{periodic}) &= \frac{1 - \text{final discount factor}}{\text{sum of discount factors}} \\ \text{swap fixed rate (annual)} &= \text{SFR}(\text{periodic}) \times \text{number of settlement periods per year} \end{aligned}$$

EXAMPLE: Calculating the fixed rate on a swap with quarterly payments

Annualized MRR spot rates today are:

$$R_{90\text{-day}} = 0.030$$

$$R_{180\text{-day}} = 0.035$$

$$R_{270\text{-day}} = 0.040$$

$$R_{360\text{-day}} = 0.045$$

You're analyzing a 1-year swap with quarterly payments and a notional principal amount of \$5,000,000. Calculate:

- The fixed rate in percentage terms.
- The quarterly fixed payments in \$.
- The first net payment.

Answer:

First calculate the discount factors. Don't forget to convert from annualized rates to per-period rates.

$$Z_{90\text{-day}} = \frac{1}{1 + \left(0.030 \times \frac{90}{360} \right)} = 0.99256$$

$$Z_{180\text{-day}} = \frac{1}{1 + \left(0.035 \times \frac{180}{360} \right)} = 0.98280$$

$$Z_{270\text{-day}} = \frac{1}{1 + \left(0.040 \times \frac{270}{360} \right)} = 0.97087$$

$$Z_{360\text{-day}} = \frac{1}{1 + \left(0.045 \times \frac{360}{360} \right)} = 0.95694$$



PROFESSOR'S NOTE

It is likely that on the exam you will be given a table of discount factors to use in pricing a swap, as in the next example. We've provided the calculations here so you know where the discount factors come from.

The quarterly fixed rate on the swap is:

$$\frac{1 - 0.95694}{0.99256 + 0.98280 + 0.97087 + 0.95694} = 0.011 = 1.1\%$$

The quarterly fixed-rate payments, assuming a notional principal of \$5,000,000, are:

$$\$5,000,000 \times 0.011 = \$55,000$$

The fixed rate on the swap in annual terms is:

$$1.1\% \times \frac{360}{90} = 4.4\%$$

We have priced a swap in which one side pays quarterly MRR and the other side pays 4.4% fixed annually (1.1% quarterly).

The first floating rate payment will be based on the 90-day rate known at inception (i.e., 3%). So the first net payment of 1.4% adjusted for quarterly time interval will be paid by the fixed-rate payer:

$$(0.014) \times (90/360) \times 5,000,000 = \$17,500$$

The amount of the first net payment will simply be the difference between the periodic fixed payment and the first floating-rate payment:

$$\$55,000 - \$17,500 = \$37,500$$

Thus, the first net payment will be \$37,500 paid by the fixed rate payer.

Calculating the Market Value of an Interest Rate Swap

For the purpose of the exam, we are only responsible for valuation of an interest rate swap *on* settlement dates (after settlement has occurred), and not *between* settlement dates.

After the initiation of an interest rate swap, the swap will take on a positive or negative value as interest rates change. The party that is the fixed-rate payer benefits if rates increase because they are paying the older (and lower) fixed rate and receiving the newer (and higher) floating rate. Similarly, the fixed-rate receiver benefits if rates decrease because they are receiving the older (and higher) fixed rate while paying a newer (and lower) floating rate. The value is calculated as present value of the difference in payments (under the new current rate, relative to the payments under the older, locked-in rate).

$$\text{value to the payer} = \sum Z \times (\text{SFR}_{\text{New}} - \text{SFR}_{\text{Old}}) \times \frac{\text{days}}{360} \times \text{notional principal}$$

where:

$\sum Z$ = the sum of discount factors associated with the remaining settlement periods

days = number of days in the settlement period

EXAMPLE: Valuing a swap on payment date

Let's continue our previous example of a one-year, quarterly settlement, \$5 million notional swap with a swap fixed rate at initiation of 4.4%. Suppose that after 180 days, the MRR term structure is flat at 3.5% over the next year. Calculate the value of the swap to the fixed-rate receiver.

Answer:

Because the MRR term structure is flat at 3.5%, the new swap fixed rate must be 3.5%. At $t = 180$, there are two more settlement dates remaining: 90 and 180 days in the future.

$$Z_{90\text{-day}} = \frac{1}{1 + \left(0.035 \times \frac{90}{360}\right)} = 0.9913$$

$$Z_{180\text{-day}} = \frac{1}{1 + \left(0.035 \times \frac{180}{360}\right)} = 0.9828$$

$$\Sigma Z = 0.9913 + 0.9828 = 1.9741$$

value of the swap to the payer =

$$1.9741 \times (0.035 - 0.044) \times \frac{90}{360} \times \$5,000,000 = -\$22,209$$

value to the receiver = +\$22,209



MODULE QUIZ 28.6

1. Consider a one-year, quarterly settlement interest rate swap. Annualized MRR spot rates and the present value factors today are the following.

	Rate	Present Value Factor
90-day MRR	4.2%	0.98961
180-day MRR	4.8%	0.97656
270-day MRR	5.0%	0.96386
360-day MRR	5.2%	<u>0.95057</u>
Total		3.88060

The annualized swap rate is *closest* to:

- A. 1.27%.
- B. 2.54%.
- C. 5.08%.

Use the following information to answer Questions 2 and 3.

Two parties enter into a 2-year fixed-for-floating interest rate swap with semiannual payments. The floating-rate payments are based on MRR. The 180-, 360-, 540-, and 720-day annualized

MRR rates and present value factors are:

	Rate	Present Value Factor
180-day MRR	5.0%	0.9756
360-day MRR	6.0%	0.9434
540-day MRR	6.5%	0.9112
720-day MRR	7.0%	0.8772

2. The swap rate is *closest* to:

- A. 6.62%.
- B. 6.87%.
- C. 7.03%.

3. After 180 days, the swap is marked-to-market when the 180-, 360-, and 540-day annualized MRRs are 4.5%, 5%, and 6%, respectively. The present value factors, respectively, are 0.9780, 0.9524, and 0.9174. What is the market value of the swap per \$1 notional principal, and which of the two counterparties (the fixed-rate payer or the fixed-rate receiver) would make the payment to mark the swap to market?

- | | |
|---------------------|------------------------|
| <u>Market value</u> | <u>Payment made by</u> |
| A. \$0.01166 | Fixed-rate payer |
| B. \$0.04290 | Fixed-rate payer |
| C. \$0.01166 | Fixed-rate receiver |

MODULE 28.7: CURRENCY SWAPS



Video covering
this content is
available online.

LOS 28.f: Describe how currency swaps are priced, and calculate and interpret their no-arbitrage value.

Currency Swaps

Determining the Fixed Rate and Foreign Notional Principal

Consider two currencies, the U.S. dollar (\$) and the British pound (£), where the exchange rate is currently \$2 per £ or £0.5 per \$. The interest rates in a currency swap are simply the swap rates calculated from each country's yield curve in the relevant country's currency. Don't forget: With currency swaps, there are two yield curves and two swap fixed rates, one for each currency.

The principal amounts of the fixed-rate obligations must be adjusted for the current exchange rate. We need \$2.00 to equal £1.00, so these are the notional amounts, exchanged at the inception of the swap and returned at the termination of the swap. For example, if the notional principal amount of the \$ side of the swap is \$25 million, the £ notional principal amount will be £12.5 million.

EXAMPLE: Calculating the fixed rate and notional principal on a currency swap

In a previous example, we determined that the fixed rate on a 1-year quarterly \$5,000,000 interest rate swap, given the following set of spot MRRs, was 1.1% quarterly, or 4.4% on an annual basis.

$$R_{90\text{-day}}^{\$} = 0.030$$

$$R_{180\text{-day}}^{\$} = 0.035$$

$$R_{270\text{-day}}^{\$} = 0.040$$

$$R_{360\text{-day}}^{\$} = 0.045$$

The comparable set of £ rates are:

$$R_{90\text{-day}}^{\pounds} = 0.04$$

$$R_{180\text{-day}}^{\pounds} = 0.05$$

$$R_{270\text{-day}}^{\pounds} = 0.06$$

$$R_{360\text{-day}}^{\pounds} = 0.07$$

The current exchange rate is £0.50 per \$.

Determine the fixed rate on a 1-year £ interest rate swap. Then determine the notional £ principal amount and the quarterly cash flows on a Pay \$ fixed, receive £ fixed currency swap.

Answer:

First calculate the £ discount factors. Don't forget to convert from annualized rates to per-period rates.

$$Z_{90\text{-day}}^{\pounds} = \frac{1}{1 + \left(0.04 \times \frac{90}{360}\right)} = 0.99010$$

$$Z_{180\text{-day}}^{\pounds} = \frac{1}{1 + \left(0.05 \times \frac{180}{360}\right)} = 0.97561$$

$$Z_{270\text{-day}}^{\pounds} = \frac{1}{1 + \left(0.06 \times \frac{270}{360}\right)} = 0.95694$$

$$Z_{360\text{-day}}^{\pounds} = \frac{1}{1 + \left(0.07 \times \frac{360}{360}\right)} = 0.93458$$

The quarterly fixed rate on the £ swap is:

$$\frac{1 - 0.93458}{0.99010 + 0.97561 + 0.95694 + 0.93458} = 0.017 = 1.7\%$$

The fixed rate on the £ swap in *annual* terms is:

$$1.7\% \times \frac{360}{90} = 6.8\%$$

The notional £ principal amount of the swap would be:

$$\$5,000,000 \times \pounds 0.50 \text{ per } \$ = \pounds 2,500,000$$

At the initiation of the swap, we would exchange £2,500,000 for \$5,000,000. We would pay 1.1% quarterly on the \$5,000,000 notional principal (\$55,000) and receive 1.7% on

£2,500,000 quarterly (£42,500). At the end of one year, we would exchange the original principal amounts.

The value to any party in a currency swap is the present value of the cash flows they expect to receive minus the present value of the cash flows they are obligated to pay.

EXAMPLE: Calculating the value of a currency swap after initiation

Use the data on the \$ and £ interest rate swaps in the previous examples to answer this question. After 300 days, the 60-day \$ interest rate is 5.4%, the 60-day £ interest rate is 6.6%, and the exchange rate is £0.52 per \$. Calculate the value of a \$5,000,000 swap to the counterparty that receives \$ fixed and pays £ fixed.

Answer:

Recall that the \$ fixed rate was calculated as 4.4% (or 1.1% per quarter) and the £ fixed rate was calculated as 6.8% (or 1.7% per quarter). After 300 days, the only cash flows remaining are the last interest payments and the principal payments in 60 days. We want to find the present value of those cash flows, so we need the 60-day discount factors based on the current 60-day rates:

$$Z_{60\text{-day}}^{\$} = \frac{1}{1 + \left(0.054 \times \frac{60}{360}\right)} = 0.99108$$

$$Z_{60\text{-day}}^{\pounds} = \frac{1}{1 + \left(0.066 \times \frac{60}{360}\right)} = 0.98912$$

First calculate the value after 300 days of the \$ fixed payments. It is equivalent to the value of a \$5,000,000 bond that matures in 60 days and makes a coupon payment in 60 days of \$55,000 (\$5,000,000 × 0.011). The value in \$ of the \$ fixed side is the present value of \$5,055,000 discounted using the 60-day \$ discount factor of 0.99108:

$$\text{value of \$ fixed side (in \$)} = 0.99108 \times \$5,055,000 = \$5,009,909$$

Next, calculate the value after 300 days of the £ fixed payments. The value in £ of the £ fixed side is the present value of £2,542,500 discounted using the 60-day £ discount factor of 0.98912:

$$\text{value of £ fixed side (in £)} = 0.98912 \times \pounds2,542,500 = \pounds2,514,838$$

This last step is to convert that amount into \$ at the *current spot exchange rate* of £0.52 per \$, to find the value of the £ fixed side in \$:

$$\text{value of £ fixed side (in \$)} = \pounds2,514,838 / 0.52 = \$4,836,227$$

Finally, the value of the receive \$ fixed, pay £ fixed side of the swap is equal to the value of the \$ fixed side minus the £ fixed side, or \$5,009,909 – \$4,836,227 = \$173,682.



MODULE QUIZ 28.7

- The current exchange rate is 0.70 USD/CAD. In a US\$1 million fixed-for-floating currency swap, the party that is entering the swap to hedge an existing exposure to a C\$-denominated fixed-rate liability will:
 - receive \$1 million at the termination of the swap.
 - pay a fixed rate based on the yield curve in the United States.
 - receive a fixed rate based on the yield curve in Canada.
- A bank entered into a 1-year currency swap with quarterly payments 200 days ago by agreeing to swap \$1,000,000 for €800,000. The bank agreed to pay an annual fixed rate of 5% on the €800,000 and receive a fixed rate of 4.2% on the \$1,000,000. Current USD and euro MRRs and present value factors are shown in the following table.

	Rate	Present Value Factor
70-day USD MRR	4.0%	0.9923
90-day USD MRR	4.4%	0.9891
160-day USD MRR	4.8%	0.9791
180-day USD MRR	5.2%	0.9747
70-day EUR MRR	5.2%	0.9900
90-day EUR MRR	5.6%	0.9862
160-day EUR MRR	6.1%	0.9736
180-day EUR MRR	6.3%	0.9695

The current spot exchange rate is €0.75 per \$1.00. The value of the swap to the bank today is *closest* to:

- \$64,888.
- \$42,049.
- \$42,049.

MODULE 28.8: EQUITY SWAPS



Video covering this content is available online.

LOS 28.g: Describe how equity swaps are priced, and calculate and interpret their no-arbitrage value.

Equity Swaps

To price an N -period pay-fixed equity swap, we can use the same formula as for a plain vanilla swap:

$$\text{SFR}(\text{periodic}) = \frac{1 - \text{final discount factor}}{\text{sum of discount factors}}$$

To value this swap, consider two scenarios:

- Valuation on settlement date (immediately after settlement): We can continue to use the valuation formula discussed for valuation of interest rate swaps.
- Valuation between settlement dates (for equity swaps, this is covered in the curriculum): We calculate the values of the two sides of the swap separately as shown in the following example.

EXAMPLE: Valuing a pay-fixed, receive-equity-returns swap

A \$10 million principal, one-year, quarterly settlement equity swap has a fixed rate of 6.05%. The index at inception is 985. After 30 days have passed, the index stands at 996 and the term structure of MRR is 6%, 6.5%, 7%, and 7.5% for terms of 60, 150, 240, and 330 days, respectively. The respective discount factors are 0.99010, 0.97363, 0.95541, and 0.93567. Calculate the value of the swap to the fixed-rate payer on Day 30.

Answer:

$$\begin{aligned} &\text{Value of the fixed side (per \$1 notional)} \\ &= \text{present value of coupons} + \text{PV of principal} \\ &= (0.0605/4) (0.99010 + 0.97363 + 0.95541 + 0.93567) + (0.93567) (1) \\ &= 0.9940 \\ &\text{Value of the fixed-rate side of the swap (\$10 million notional)} \\ &= 0.9940 \times \$10 \text{ million} = \$9,940,000 \end{aligned}$$

The value of \$10,000,000 invested in the index after 30 days is:

$$\$10,000,000 \times (996/985) = \$10,111,675$$

From the standpoint of the fixed-rate payer, the value of the swap after 30 days is:

$$\$10,111,675 - \$9,940,000 = \$171,675.$$

A swap of returns on two different stocks can be viewed as buying one stock (receiving the returns) and shorting an equal value of a different stock (paying the returns). There is no “pricing” at swap initiation, and we can value the swap at any point in time by taking the difference in returns (since the last payment date) times the notional principal.

EXAMPLE: Valuing a “one-equity-return-for-another” swap

An investor is the Stock A returns payer (and Stock B returns receiver) in a \$1 million quarterly-pay swap. After one month, Stock A is up 1.3% and Stock B is down 0.8%. Calculate the value of the swap to the investor.

Answer:

The investor pays the Stock A returns and receives Stock B returns. However, the Stock B returns are negative, so he pays those as well:

$$\text{value of swap} = (-0.013 - 0.008) \times \$1,000,000 = -\$21,000$$

**MODULE QUIZ 28.8**

1. A bank entered into a \$5,000,000, 1-year equity swap with quarterly payments 300 days ago. The bank agreed to pay an annual fixed rate of 4% and receive the return on an international equity index. The index was trading at 3,000 at the end of the third quarter, 30 days ago. The current 60-day MRR rate is 3.6%, the discount factor is 0.9940, and the index is now at 3,150. The value of the swap to the bank is *closest* to:
A. -\$257,795.
B. -\$114,676.

KEY CONCEPTS

LOS 28.a

The calculation of the forward price for an equity forward contract is different because the periodic dividend payments affect the no-arbitrage price calculation. The forward price is reduced by the future value of the expected dividend payments; alternatively, the spot price is reduced by the present value of the dividends.

$FP(\text{on an equity security}) =$

$$(S_0 - PVD) \times (1 + R_f)^T = [S_0 \times (1 + R_f)^T] - FVD$$

The value of an equity forward contract to the long is the spot equity price minus the present value of the forward price minus the present value of any dividends expected over the term of the contract:

$$V_t(\text{long position}) = [S_t - PVD_t] - \left[\frac{FP}{(1 + R_f)^{(T-t)}} \right]$$

If given the current forward price (FP_t) on the same underlying and with the same maturity:

$$V_t(\text{long position}) = \left[\frac{FP_t - FP}{(1 + R_f)^t} \right]$$

We typically use the continuous time versions to calculate the price and value of a forward contract on an equity index using a continuously compounded dividend yield.

$$FP(\text{on an equity index}) = S_0 \times e^{(R_f - \delta^c) \times T} = (S_0 \times e^{-\delta^c \times T}) \times e^{R_f \times T}$$

LOS 28.b

Forward price = spot price + net cost of carry

For a security without underlying cash flows:

$$FP = S_0 (1 + R_f)^T$$

For a security with underlying cash flows:

$$FP = (S_0 - PVC) \times (1 + R_f)^T$$

where PVC = present value of the cash flow on the security.

LOS 28.c

The “price” of an FRA is the implied forward rate for the period beginning when the FRA expires to the maturity of the underlying “loan.”

The value of an FRA at maturity is the interest savings to be realized at maturity of the underlying “loan” discounted back to the date of the expiration of the FRA at the current

MRR. The value of an FRA before maturity is the interest savings estimated by the implied forward rate discounted back to the valuation date at the current MRR.

LOS 28.d

For forwards on coupon-paying bonds, the price is calculated as the spot price minus the present value of the coupons times the quantity one plus the risk-free rate:

$$FP \text{ (on a fixed income security)} = (S_0 - PVC) \times (1 + R_f)^T = S_0 \times (1 + R_f)^T - FVC$$

The value of a forward on a coupon-paying bond t years after inception is the spot bond price minus the present value of the forward price minus the present value of any coupon payments expected over the term of the contract:

$$V_t(\text{long position}) = [S_t - PVC_t] - \left[\frac{FP}{(1 + R_f)^{(T-t)}} \right]$$

If given the current forward price (FP_t) on the same underlying and with the same maturity:

$$V_t(\text{long position}) = \left[\frac{FP_t - FP}{(1 + R_f)^t} \right]$$

In a futures contract, the short may have delivery options (to decide which bond to deliver). In such a case, the quoted futures price is adjusted using the conversion factor for the cheapest-to-deliver bond:

$$QFP = FP / CF = [(full \text{ price})(1 + R_f)^T - AL_T - FVC] \left(\frac{1}{CF} \right)$$

where:

the full price = clean price + accrued interest at $t = 0$

AL_T = accrued interest at futures contract maturity.

LOS 28.e

The fixed periodic-rate on an n -period swap at initiation (as a percentage of the principal value) can be calculated as:

$$SFR(\text{periodic}) = \frac{1 - \text{final discount factor}}{\text{sum of discount factors}}$$

where:

$$\text{discount factor} = Z = \frac{1}{\left[1 + \left(MRR \times \frac{\text{days}}{360} \right) \right]}$$

The value of a swap on a payment date has a simple relationship to the difference between the new swap fixed rate and the original swap fixed rate:

$$\text{value to the payer} = \Sigma Z \times (SFR_{\text{New}} - SFR_{\text{Old}}) \times \frac{\text{days}}{360} \times \text{notional principal}$$

where:

ΣZ = sum of discount factors associated with the *remaining* settlement periods

LOS 28.f

The fixed rates in a fixed-for-fixed currency swap are determined using the yield curves for the relevant currencies. The notional principal amounts in the two currencies are of equal value, based on the exchange rate at inception of the swap. Use the difference in values to value the currency swap. The conversion of value from one of the two currencies into the common currency is based on the exchange rate on the valuation date.

LOS 28.g

The fixed-rate side of an equity swap is priced and valued just like an interest rate swap. The equity side can be valued by multiplying the notional amount of the contract by $1 +$ the percentage equity appreciation since the last payment date. Use the difference in values to value the swap.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 28.1, 28.2

1. **C** The dividend in 65 days occurs after the contract has matured, so it's not relevant to computing the forward price.

$$\begin{aligned} \text{PVD} &= \frac{\$0.30}{1.05^{20/365}} = \$0.2992 \\ \text{FP} &= (\$30.00 - \$0.2992) \times 1.05^{60/365} = \$29.94 \\ &\text{(Module 28.2, LOS 28.a)} \end{aligned}$$

2. **B**

$$\begin{aligned} V(\text{long position}) &= \$21.00 - \left[\frac{\$29.94}{1.05^{23/365}} \right] = -\$8.85 \\ V(\text{short position}) &= +\$8.85 \\ &\text{(Module 28.2, LOS 28.a)} \end{aligned}$$

3. **B** Use the dividend rate as a continuously compounded rate to get:

$$\begin{aligned} \text{FP} &= 540 \times e^{(0.07 - 0.018) \times (200 / 365)} = 555.61 \\ &\text{(Module 28.2, LOS 28.a)} \end{aligned}$$

4. **B** The value of the long position in a forward contract on a stock at time t is:

$$V_t(\text{long position}) = [S_t - \text{PVD}_t] - \left[\frac{\text{FP}}{(1 + R_f)^{(T-t)}} \right]$$

If the dividends are ignored, the *long* position will be overvalued by the present value of the dividends; that means the *short* position (which is what the question asks for) will be undervalued by the same amount. (Module 28.2, LOS 28.a)

5. **A** The dividend in 125 days is irrelevant because it occurs after the forward contract matures.

$$PVD = \frac{\$0.50}{1.04^{35/365}} = \$0.4981$$

$$FP = (\$35 - \$0.4981) \times 1.04^{120/365} = \$34.95$$

$$V_{45}(\text{short position}) = -\left(\$27.50 - \frac{\$34.95}{1.04^{75/365}}\right) = \$7.17$$

(Module 28.2, LOS 28.a)

Module Quiz 28.3

1. **B** Remember that U.S. Treasury bonds make semiannual coupon payments, so:

$$C = \frac{\$1,000 \times 0.06}{2} = \$30.00$$

$$PVC = \frac{\$30.00}{1.05^{98/365}} = \$29.61$$

The forward price of the contract is therefore:

$$\begin{aligned} &FP \text{ (on a fixed income security)} \\ &= (S_0 - PVC) \times (1 + R_f)^T = (\$1,044 - \$29.61) \times (1.05)^{120/365} \\ &= \$1,030.79 \end{aligned}$$

(LOS 28.d)

Module Quiz 28.4

1. **C** The actual (unannualized) rate on the 90-day loan is:

$$R_{90} = 0.05 \times \frac{90}{360} = 0.0125$$

The actual rate on the 150-day loan is:

$$R_{150} = 0.06 \times \frac{150}{360} = 0.025$$

The price of the 3 × 5 FRA (the 60-day forward rate in 90 days) is:

$$\left(\frac{1.025}{1.0125} - 1\right) \times \frac{360}{60} = 0.074 = 7.4\%$$

(LOS 28.c)

2. **C** A 3 × 6 FRA expires in 90 days and is based on 90-day MRR, so it is the appropriate hedge for 90-day MRR 90 days from today. The rate is calculated as:

$$R_{90} = 0.047 \times \frac{90}{360} = 0.0118$$

$$R_{180} = 0.049 \times \frac{180}{360} = 0.0245$$

$$\text{price of } 3 \times 6 \text{ FRA} = \left(\frac{1.0245}{1.0118} - 1\right) \times \frac{360}{90} = 0.0502 = 5.02\%$$

(LOS 28.c)

Module Quiz 28.5

1. **A** The original FRA will now be a 2×5 FRA (as 1 month has passed).

$$\begin{aligned}
 \text{new FRA} &= \left\{ \left[\frac{(1 + L150 \times 150/360)}{(1 + L60 \times 60/360)} - 1 \right] \times \left[\frac{360}{90} \right] \right\} \\
 &= \left\{ \left[\frac{(1 + 0.032 \times 150/360)}{(1 + 0.026 \times 60/360)} - 1 \right] \times 4 \right\} \\
 &= \left[\frac{(1.01333}{1.00433)} - 1 \right] \times 4 \\
 &= 3.584\% \\
 \text{value (long)} &= \text{PV}[(0.03584 - 0.03) \times (90/360) \times 10\text{m}] \\
 &= \text{PV}(14,600) \\
 &= (14,600) / (1 + 0.032 \times 150/360) \\
 &= 14,600 / 1.01333 = 14,407
 \end{aligned}$$

(LOS 28.c)

Module Quiz 28.6

1. **C** The quarterly fixed rate on the swap is:

$$\frac{1 - 0.95057}{3.88060} = 0.0127 = 1.27\%$$

The fixed rate on the swap in annual terms is:

$$1.27\% \times \frac{360}{90} = 5.08\%$$

(LOS 28.e)

2. **A** Calculate the swap rate:

$$\text{semi-annual swap rate} = \frac{1 - 0.8772}{0.9756 + 0.9434 + 0.9112 + 0.8772} = 0.0331$$

$$\text{swap rate} = 0.0331 \times \frac{360}{180} = 0.0662 = 6.62\%$$

(LOS 28.e)

3. **A** Based on new rates, $\Sigma DF = 0.9780 + 0.9524 + 0.9174 = 2.8478$.

$$\text{new SFR} = \frac{1 - 0.9174}{2.8478} \times \frac{360}{180} = 0.0580$$

$$\text{value (payer)} = \frac{(0.0580 - 0.0662)}{2} \times 2.8478 = -0.01166 \text{ per \$1 notional}$$

Because the value is negative, payer makes the payment. (LOS 28.e)

Module Quiz 28.7

1. **C** A receive-fixed C\$ position will hedge the liability risk. That party would receive \$1 million at swap inception (in exchange for $\frac{1,000,000}{0.7} = \text{C\$1,428,571}$) and pay it back at termination. The fixed-rate received will be calculated using the yield curve in Canada at the initiation of the swap. Because this is a fixed-for-floating currency swap, the receive-fixed position will pay a floating rate based on the U.S. yield curve. (LOS 28.f)

2. **A** coupon on \$ fixed side = $\$1,000,000 \times (0.042 / 4) = \$10,500$

$$\text{value of the \$ fixed side} = (0.9923 \times \$10,500) + (0.9791 \times \$1,010,500) \\ = \$999,800$$

$$\text{coupon on € fixed side} = €800,000 \times (0.05 / 4) = €10,000$$

$$\text{value of € fixed side (in €)} = (0.9900 \times €10,000) + (0.9736 \times €810,000) \\ = €798,516$$

$$\text{value of € fixed side (in \$)} = \frac{€798,516}{0.75} = \$1,064,688$$

$$\text{value of swap to bank} = \$999,800 - \$1,064,688 = -\$64,888 \\ (\text{LOS 28.f})$$

Module Quiz 28.8

1. **C** The quarterly 4% fixed-rate payment in 60 days will be in the amount of: $\$5,000,000 \times (4\%/4) = \$50,000$

So the total cash flow at that time is $\$50,000 + \$5,000,000 = \$5,050,000$.

$$\text{value of fixed-rate side} = 0.9940 \times \$5,050,000 = \$5,019,700$$

$$\text{value of index return side} = \$5,000,000 \times \frac{3,150}{3,000} = \$5,250,000$$

$$\text{value of swap to bank} = \$5,250,000 - \$5,019,700 = \$230,300$$

(LOS 28.g)

READING 29

VALUATION OF CONTINGENT CLAIMS

EXAM FOCUS

This topic review covers the valuation of options. Candidates need to be able to calculate value of an option using the binomial tree framework and should also understand the inputs into the Black-Scholes model and how they influence the value of an option. While this topic review is somewhat quantitative, candidates need to understand the material conceptually as well. This reading has a lot of testable material.



PROFESSOR'S NOTE

We cover these three LOS here together, for ease of exposition.

MODULE 29.1: THE BINOMIAL MODEL



Video covering this content is available online.

LOS 29.a: Describe and interpret the binomial option valuation model and its component terms.

LOS 29.b: Describe how the value of a European option can be analyzed as the present value of the option's expected payoff at expiration.

LOS 29.d: Calculate the no-arbitrage values of European and American options using a two-period binomial model.

Binomial Model

A **binomial model** is based on the idea that, over the next period, the value of an asset will change to one of two possible values (binomial). To construct a binomial model, we need to know the beginning asset value, the size of the two possible changes, and the probability of each of these changes occurring.

One-Period Binomial Model

Consider a share of stock currently priced at \$30. Suppose also that the size of the possible price changes, and the probability of these changes occurring are as follows:

S_0 = current stock price = \$30

U = up move factor = $(1 + \% \text{ up}) = S^+/S = 1.333$

D = down move factor = $(1 - \% \text{ down}) = S^-/S = 0.75$

S^+ = stock price if an up move occurs = $S_0 \times U = 30 \times 1.333 = \40

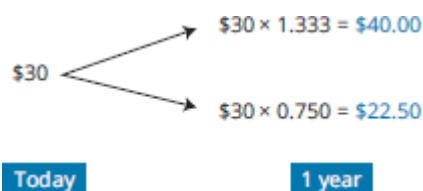
S^- = stock price if a down move occurs = $S_0 \times D = 30 \times 0.75 = \22.50

π_U = probability of an up move

π_D = probability of a down move = $(1 - \pi_U)$

A one-period binomial tree for this stock is shown in Figure 29.1. The beginning stock value of \$30 is to the left, and to the right are the two possible paths the stock can take, based on that starting point and the size of an up- or down-move. If the stock price increases by a factor of 1.333 (a return of 33.3%), it ends up at \$40.00; if it falls by a factor of 0.75 (a return of -25%), it ends up at \$22.50.

Figure 29.1: One-Period Binomial Tree



The probabilities of an up-move and a down-move are calculated based on the size of the moves, as well as the risk-free rate, as:

$$\pi_U = \frac{1 + R_f - D}{U - D}$$

where:

R_f = periodically compounded annual risk-free rate



PROFESSOR'S NOTE

These up- and down-move probabilities are not the *actual* probabilities of up- or down-moves. They are the *risk-neutral* probabilities that are consistent with investor risk-neutrality. The distinction between actual probabilities and risk-neutral probabilities is not relevant for the exam.

We use the **expectations approach** to calculate the value of an option from this binomial tree. We do this by:

- Calculating the payoff of the option at maturity in both the up-move and down-move states.
- Calculating the expected value of the option in one year as the probability-weighted average of the payoffs in each state.
- Discounting the expected value back to today at the risk-free rate.

EXAMPLE: Calculating call option value with a one-period binomial tree

Calculate the value today of a one-year call option on a stock that has an exercise price of \$30. Assume that the periodically compounded (as opposed to continuously compounded) risk-free rate is 7%, the current value of the stock is \$30, the up-move factor is 1.333, and the down move factor is 0.75.

Answer:

First, we have to calculate the probabilities:

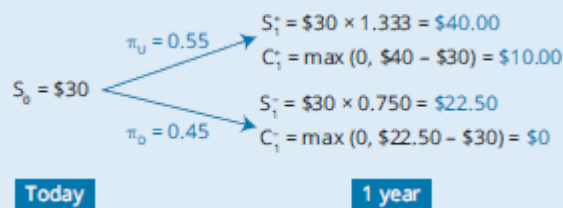
$$\pi_U = \frac{1 + R_f - D}{U - D} = \frac{1 + 0.07 - 0.75}{1.33 - 0.75} = 0.55 = 55\%$$

$$\pi_D = 1 - 0.55 = 0.45 = 45\%$$

Next, determine the option payoffs in each state. If the stock moves up to \$40, the call option with an exercise price of \$30 will pay off \$10. If the stock moves down to \$22.50, the call option will be worthless. The option payoffs are illustrated in the following figure.

Let the stock values for the up-move and down-move be S^+ and S^- and for the call values, C^+ and C^- .

With X = \$30



The expected value of the option in one period is:

$$E(\text{call option value in 1 year}) = (\$10 \times 0.55) + (\$0 \times 0.45) = \$5.50$$

The value of the option today, after discounting at the risk-free rate of 7%, is:

$$C = \frac{\$5.50}{1.07} = \$5.14$$

We can use the same framework to value a one-period put option on the stock.

EXAMPLE: Valuing a one-period put option on a stock

Use the information in the previous example to calculate the value of a put option on the stock that has an exercise price of \$30.

Answer:

If the stock moves up to \$40, a put option with an exercise price of \$30 will be worthless. If the stock moves down to \$22.50, the put option will be worth \$7.50.

The probabilities are 0.55 and 0.45 for an up- and down-move, respectively. The expected value of the put option in one period is thus:

$$E(\text{put option value in 1 year}) = (\$0 \times 0.55) + (\$7.50 \times 0.45) = \$3.375$$

The value of the option today, after discounting at the risk-free rate of 7%, is:

$$P = \frac{\$3.375}{1.07} = \$3.154$$

MODULE 29.2: TWO PERIOD BINOMIAL MODEL AND PUT-CALL PARITY



Video covering this content is available online.

Put-Call Parity

Put option value can also be computed using put-call parity, which recognizes that the value of a fiduciary call (long call, plus an investment in a zero-coupon bond with a face value equal to the strike price) is equal to the value of a protective put (long stock and long put):

$$S_0 + P_0 = C_0 + PV(X)$$

Note that both options are on the same underlying stock have the same exercise price, and the same maturity.

In our previous example, $PV(X) = 30 / 1.07 = \$28.04$. We can verify the put call parity as:

$$S_0 + P_0 = \$30 + \$3.15 = \$33.15$$

$$C_0 + PV(X) = \$5.14 + 28.04 = \$33.19 \text{ (rounding accounts for the slight difference)}$$

Put call parity can be used to create a synthetic instrument that replicates the desired instrument.

For example, a synthetic call can be created by creating a portfolio that combines a long position in the stock, a long position in a put and a short position in a zero-coupon bond with a face value equal to the strike price (i.e., borrowing the present value of the exercise price at the risk-free rate).

$$C_0 = S_0 + P_0 - PV(X)$$

EXAMPLE: Using put-call parity

A 1-year call option on the common stock of Cross Reef, Inc., with an exercise price of \$60 is trading for \$8. The current stock price is \$62. The risk-free rate is 4%. Calculate the price of the put option implied by put-call parity.

Answer:

According to put-call parity, to prevent arbitrage, the price of the put option must be:

$$\begin{aligned} P_0 &= C_0 - S_0 + \left[\frac{X}{(1 + R_f)^T} \right] \\ &= \$8 - \$62 + \frac{\$60}{1.04} \\ &= \$3.69 \end{aligned}$$

Two-Period Binomial Model

Valuing an option using a two-period binomial model requires more steps, but uses the same method:

- Calculate the stock values at the end of two periods (there are three possible outcomes, because an up-then-down move gets you to the same place as a down-then-up move).
- Calculate the three possible option payoffs at the end of two periods.
- Calculate the expected option payoff at the end of two periods ($t = 2$) using the up- and down-move probabilities.
- Discount the expected option payoff ($t = 2$) back one period at the risk-free rate to find the option values at the end of the first period ($t = 1$).
- Calculate the expected option value at the end of one period ($t = 1$) using up- and down-move probabilities.
- Discount the expected option value at the end of one period ($t = 1$) back one period at the risk-free rate to find the option value today ($t = 0$).

Let's look at an example to illustrate the steps involved.

EXAMPLE: Valuing a call option on a stock with a two-period model

Suppose you own a stock currently priced at \$50 and that a two-period European call option on the stock is available with a strike price of \$45. The up-move factor is 1.25 and the down-move factor is 0.80. The risk-free rate per period is 7%. Compute the value of the call option using a two-period binomial model.

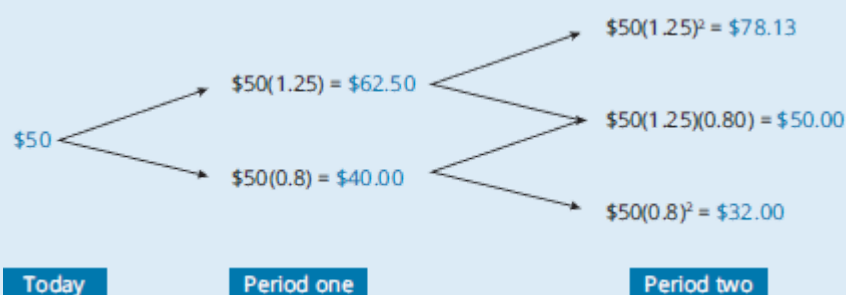
Answer:

First, compute the probability of an up-move and a down-move, and then compute the theoretical value of the stock at the end of each period:

$$\pi_U = \frac{1 + R_f - D}{U - D} = \frac{1 + 0.07 - 0.80}{1.25 - 0.80} = 0.60$$
$$\pi_D = 1 - 0.60 = 0.40$$

The two-period binomial tree for the stock is shown in the following figure.

Two-Period Binomial Tree for Stock Price



We know the value of the option at expiration in each state is equal to the stock price minus the exercise price (or zero, if that difference is negative):

$$\begin{aligned}
C^{++} &= \max(0, \$78.13 - \$45.00) = \$33.13 \\
C^{-+} &= \max(0, \$50.00 - \$45.00) = \$5.00 \\
C^{+-} &= \max(0, \$50.00 - \$45.00) = \$5.00 \\
C^{--} &= \max(0, \$32.00 - \$45.00) = \$0
\end{aligned}$$

We will approach this problem by using the single-period binomial model for each period. Using this method, we can compute the value of the call option in the up-state in period one as follows:

$$\begin{aligned}
C^+ &= \frac{E(\text{call option value})}{1 + R_f} = \frac{(\pi_U \times C^{++}) + (\pi_D \times C^{-+})}{1 + R_f} \\
&= \frac{(0.6 \times \$33.13) + (0.4 \times \$5.00)}{1.07} \\
&= \frac{\$21.88}{1.07} = \$20.45
\end{aligned}$$

The value of the call in the down-state at $t = 1$ is computed as:

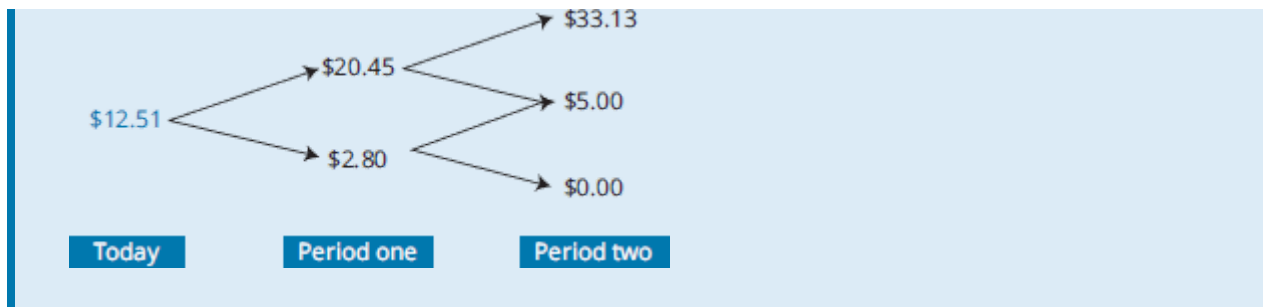
$$\begin{aligned}
C^- &= \frac{E(\text{call option value})}{1 + R_f} = \frac{(\pi_U \times C^{-+}) + (\pi_D \times C^{--})}{1 + R_f} \\
&= \frac{(0.6 \times \$5.00) + (0.4 \times \$0.00)}{1.07} \\
&= \frac{\$3.00}{1.07} = \$2.80
\end{aligned}$$

Now we know the value of the option in both the up-state (C^+) and the down-state (C^-) one period from now. To get the value of the option today, we simply apply our methodology one more time. Therefore, bringing (C^+) and (C^-) back one more period to the present, the value of the call option today is:

$$\begin{aligned}
C &= \frac{E(\text{call option value})}{1 + R_f} = \frac{(\pi_U \times C^+) + (\pi_D \times C^-)}{1 + R_f} \\
&= \frac{(0.6 \times \$20.45) + (0.4 \times \$2.80)}{1.07} \\
&= \frac{\$13.39}{1.07} = \$12.51
\end{aligned}$$

The binomial tree for the call option is shown here:

Two-Period Binomial Tree for Option Price



MODULE QUIZ 29.1, 29.2

1. Zepo, Inc., stock price is currently \$80. The stock price will move up by 15% or down by 13% each period. The value of a two-period European call option with an exercise price of \$62 if the risk-free rate is 4% per period is *closest* to:
 - A. \$19.17.
 - B. \$22.99.
 - C. \$27.11.

MODULE 29.3: AMERICAN OPTIONS



Video covering this content is available online.

American-Style Options

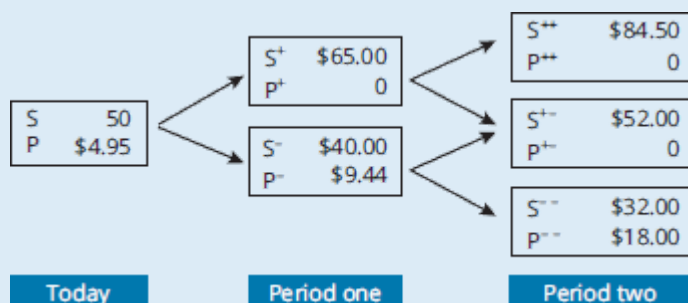
Our discussion so far has been limited to European-style options (i.e., options that can only be exercised at maturity). American-style options allow for exercise any point until maturity of the option. While the early exercise feature is not valuable for call options on non-dividend paying stocks, deep-in-the-money put options could benefit from early exercise. When an investor exercises an option early, she captures only the intrinsic value of the option and forgoes the time value. While the intrinsic value can be invested at the risk-free rate, the interest so earned is less than the time value in most cases. For a deep-in-the-money put option, the upside is limited (because the stock price cannot fall below zero). In such cases, the interest on intrinsic value can exceed the option's time value.

EXAMPLE: Early exercise of a put option

Consider a stock currently trading at \$50. The periodically compounded interest rate is 3%. Suppose that $U = 1.3$ and $D = 0.80$. Calculate the value of a two-period European-style put option on the stock that has an exercise price of \$50. Also, determine if early exercise would make economic sense.

Answer:

The following tree shows put option value at each of the nodes in the binomial tree.



The value of the option for the down node at $t = 1$ is calculated as:

$$P^- = \frac{(\pi_U \times P^{+-} + \pi_D \times P^{--})}{(1 + R_f)} = \frac{(0.46 \times 0 + 0.54 \times 18)}{(1.03)} = \$9.44$$

The value of the option at time = 0 is calculated as:

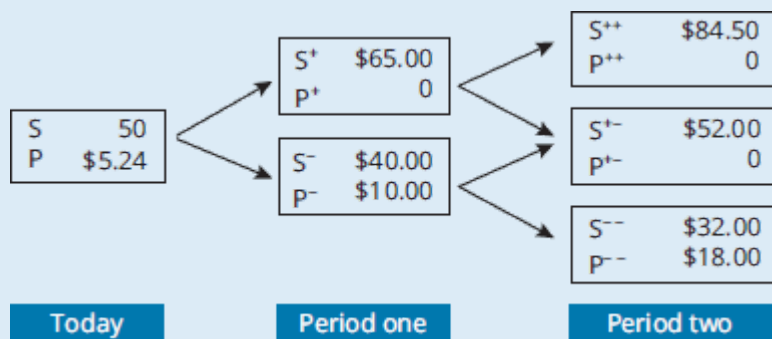
$$P = \frac{(\pi_U \times P^+ + \pi_D \times P^-)}{(1 + R_f)} = \frac{(0.46 \times 0 + 0.54 \times 9.44)}{(1.03)} = \$4.95$$

For the down move at $t = 1$, the exercise value of the put option is \$10, calculated as $\text{Max}(0, X - S)$ or $\text{Max}(0, 50 - 40)$. Clearly, for this node, early exercise results in higher value (\$10 as opposed to \$9.44).

Had the option in the previous example been an American-style put option, the value would be \$5.24 as shown in Figure 29.2.

Figure 29.2: Valuing an American-Style Put Option

The value of the put option at time 0 can be calculated as the present value of the expected value of the option at time $t = 1$.



$$P_0 = \frac{(\pi_U \times P^+ + \pi_D \times P^-)}{(1 + R_f)} = \frac{(0.46 \times 0 + 0.54 \times 10.0)}{(1.03)} = \$5.24$$

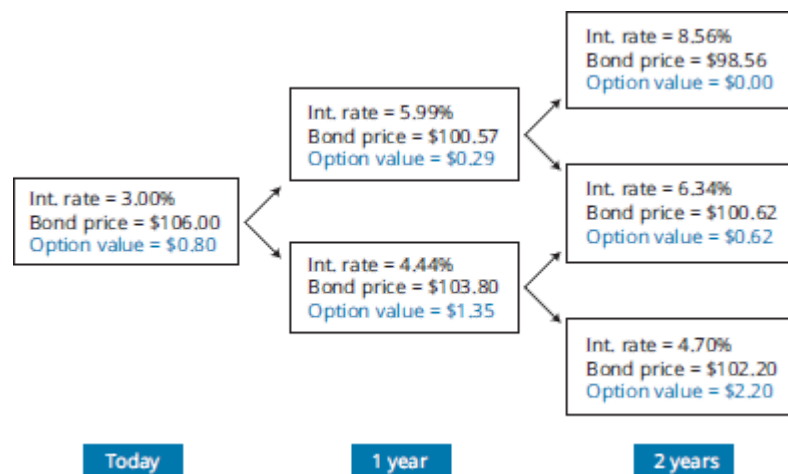
American-style call options on dividend-paying stocks can be evaluated similarly: determine at each node whether the exercise value is greater than the intrinsic value and, if so, use that higher value. For dividend-paying stocks, the stock price falls when the stock goes ex-dividend, and it may make sense to exercise the call option before such a decline in price.



MODULE QUIZ 29.3

1. An analyst has calculated the value of a 2-year European call option to be \$0.80. The strike price of the option is 100.00, and the underlying asset is a 7% annual coupon bond with three years to maturity. The two-period binomial tree for the European option is shown in

the following figure.



The value of the comparable 2-year American call option (exercisable after 1 year) with a strike price of 100.00 is *closest* to:

- A. \$1.56.
- B. \$2.12.
- C. \$3.80.

MODULE 29.4: HEDGE RATIO



Video covering this content is available online.

LOS 29.c: Identify an arbitrage opportunity involving options and describe the related arbitrage.

Arbitrage with a One-Period Binomial Model

Let's revisit our original example of a single period binomial model. Recall that the call option has a strike price equal to the stock's current price of \$30, $U = 1.333$, $D = 0.75$, and the risk-free rate is 7%. We calculated that the probability of an up-move is 55% and that of a down-move 45%.

If the market price of the one-period \$30 call option were to be different from the \$5.14 value calculated before, there would be an arbitrage opportunity. This arbitrage will involve the call option and shares of the stock. If the option is overpriced in the market, we would sell the option and buy a fractional share of the stock for each option we sold. If the call option is underpriced in the market, we could purchase the option and short a fractional share of stock for each option purchased.

The fractional share of stock needed in the arbitrage trade (commonly referred to as the **hedge ratio** or delta), is calculated in the one-period model as:

$$h = \frac{C^+ - C^-}{S^+ - S^-}$$

EXAMPLE: Calculating arbitrage profit

In a previous example, a one-year call option on a \$30 stock (exercise price = \$30) was valued at \$5.14. We were given that the risk-free rate is 7%, and the up-move factor and down-move factor are 1.333 and 0.75, respectively. If the market price of the call option

is \$6.50, illustrate how this opportunity can be exploited to earn an arbitrage profit. Assume we trade 100 call options.

Answer:

Given the up and down factors, the stock price would be \$40 and \$22.50 in the up and down states respectively in one year. The call payoffs would accordingly be \$10 and \$0 in up and down states, respectively. Because the option is overpriced, we will sell 100 call options and purchase a number of shares of stock determined by the hedge ratio:

$$h = \frac{\$10 - \$0}{\$40 - \$22.50} = 0.5714 \text{ shares per option}$$

$$\text{Total number of shares to purchase} = 100 \times 0.5714 = 57.14$$

A portfolio that is long 57.14 shares of stock at \$30 per share and short 100 calls at \$6.50 per call has a net cost of:

$$\text{net portfolio cost} = (57.14 \times \$30) - (100 \times \$6.50) = \$1,064$$

We will borrow \$1,064 at 7%, and then will have to repay $\$1,064 \times (1.07) = \$1,138.48$ at the end of one year. (In an arbitrage transaction, we assume that we begin with \$0.)

The values of this portfolio at maturity if the stock moves up to \$40 or down to \$22.50 are identical:

$$\text{portfolio value after up-move} = (57.14 \times \$40) - (100 \times \$10) = \$1,286$$

$$\text{portfolio value after down-move} = (57.14 \times \$22.50) - (100 \times \$0) = \$1,286$$

Profit on the portfolio at the end of one year in either state after repayment of loan = $\$1,286 - \$1,138.48 = \$147.52$.

The present value of the arbitrage profit is $\$147.52 / 1.07 = \137.87 .

The previous calculations can be represented by the following:

$$-hS_0 + C_0 = \frac{(-hS^- + C^-)}{(1 + R_f)}$$

Therefore, the value of the call (the **arbitrage-free price**) $C_0 = hS_0 + \frac{(-hS^- + C^-)}{(1 + R_f)}$

Or equivalently,

$$C_0 = hS_0 + \frac{(-hS^+ + C^+)}{(1 + R_f)}$$

We can value the call option using the first relationship as:

$$C_0 = (0.5714 \times \$30) + \frac{(-0.5714 \times \$22.50) + \$0}{(1.07)}$$

= \$5.12 (the small difference is due to rounding)

Similarly, a put option can be valued as:

$$P_0 = hS_0 + \frac{(-hS^- + P^-)}{(1 + R_f)} = hS_0 + \frac{(-hS^+ + P^+)}{(1 + R_f)}$$

And the hedge ratio based on puts is:

$$h = \frac{p^+ - p^-}{s^+ - s^-}$$



MODULE QUIZ 29.4

1. In a one-period binomial model, the hedge ratio is 0.35. To construct a riskless arbitrage involving 1,000 call options if the option is “overpriced,” what is the appropriate portfolio?

- | | <u>Calls</u> | <u>Stock</u> |
|----|--------------------|--------------------|
| A. | Buy 1,000 options | Short 350 shares |
| B. | Buy 1,000 options | Short 2,857 shares |
| C. | Sell 1,000 options | Buy 350 shares |

2. A synthetic European put option is created by:

- A. buying the discount bond, buying the call option, and short-selling the stock.
- B. buying the call option, short-selling the discount bond, and short-selling the stock.
- C. short-selling the stock, buying the discount bond, and selling the call option.

MODULE 29.5: INTEREST RATE OPTIONS



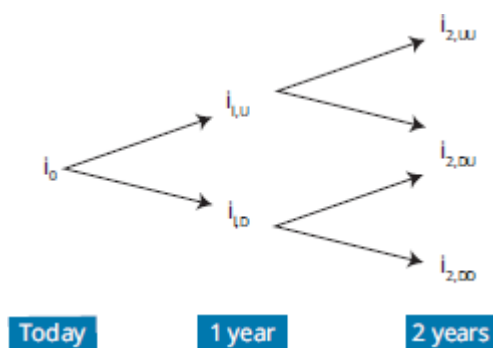
Video covering this content is available online.

LOS 29.e: Calculate and interpret the value of an interest rate option using a two-period binomial model.

Warm-Up: Binomial Interest Rate Trees

We can use an estimate of the volatility of an interest rate to create a set of possible rate paths for interest rates in the future called a **binomial interest rate tree**. The diagram in Figure 29.3 depicts a two-period binomial interest rate tree.

Figure 29.3: Two-Period Binomial Interest Rate Tree



The interest rate at each node is a one-period forward rate. Beyond the root of the tree, there is more than one one-period forward rate for each nodal period (i.e., at Year 1, we have two 1-year forward rates, $i_{1,U}$ and $i_{1,D}$). The interest rates are selected so that the (risk-neutral) probabilities of up- and down-moves are both equal to 0.5.



PROFESSOR'S NOTE

You will not have to construct a tree for the exam—just be able to use the given tree to value an interest rate option.

Interest Rate Options

An interest rate call option has a positive payoff when the reference rate is greater than the exercise rate:

$$\text{call payoff} = \text{notional principal} \times [\text{Max}(0, \text{reference rate} - \text{exercise rate})]$$

Interest rate call options increase in value when rates increase.

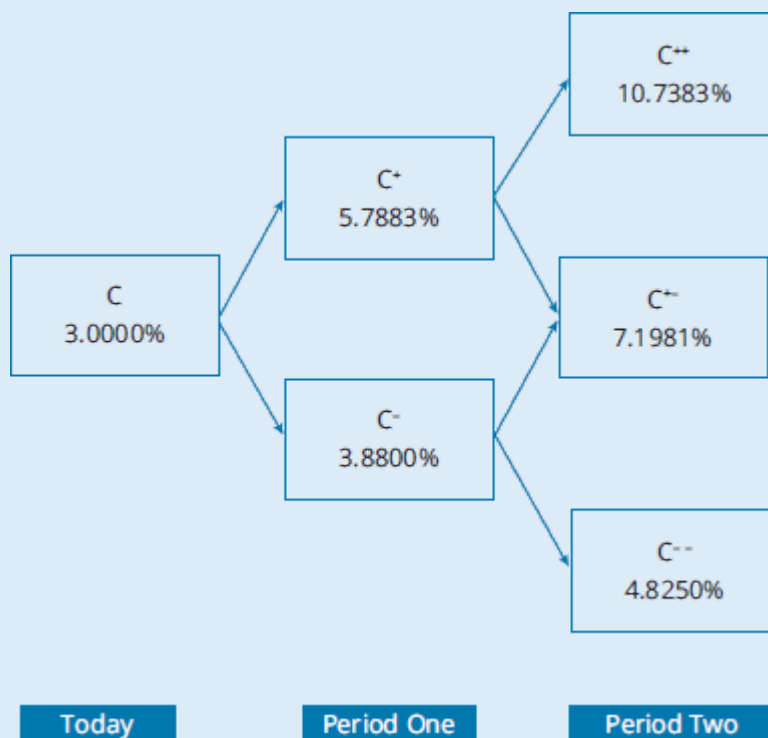
An interest rate put option has a positive payoff when the reference rate is less than the exercise rate:

$$\text{put payoff} = \text{notional principal} \times [\text{Max}(0, \text{exercise rate} - \text{reference rate})]$$

Interest rate put option values increase in value when rates decrease. Valuing interest rate options using the binomial tree is similar to valuing stock options; the value at each node is the present value of the expected value of the option. While MRR-based contracts pay interest in arrears, to keep things simple, we assume that options cash settle at maturity.

EXAMPLE: Interest rate call option valuation

Given the two-period interest rate tree shown next, what is the value of a two-period European interest rate call option with an exercise rate of 5.50% and a notional principal of \$1 million? (Assume that options cash settle at time $T = 2$.)



Answer:

Given the exercise rate of 5.50%, the call option has a positive payoff for nodes C^{++} and C^{+-} .

The value of the option at node C^{++} can be calculated as:

$$[\text{Max}(0, 0.107383 - 0.055)] \times \$1,000,000 = \$52,383$$

Similarly, the value at node C^{+-} can be calculated as:

$$[\text{Max}(0, 0.071981 - 0.055)] \times \$1,000,000 = \$16,981$$

value at node $C^+ = [(0.5 \times 52,383) + (0.5 \times 16,981)] / (1.057883) = \$32,784$ (note that the discount rate is not constant)

value at node $C^- = [(0.5 \times 16,981) + 0] / (1.0388) = \$8,173$

value at node $C = [(0.5 \times 32,784) + (0.5 \times 8,173)] / (1.03) = \$19,882$

MODULE 29.6: BLACK-SCHOLES-MERTON AND SWAPTIONS



Video covering
this content is
available online.

LOS 29.f: Identify assumptions of the Black–Scholes–Merton option valuation model.

The **Black-Scholes-Merton (BSM)** option valuation **model** values options in continuous time, but is based on the no-arbitrage condition we used in valuing options in discrete time with a binomial model. In the binomial model, the hedge portfolio is riskless over the next period, and the no-arbitrage option price is the one that guarantees that the hedge portfolio will yield the risk-free rate. To derive the BSM model, an “instantaneously” riskless portfolio (one that is riskless over the next instant) is used to solve for the option price.

The assumptions underlying the Black-Scholes-Merton model are:

1. The underlying asset price follows a geometric Brownian motion process. The continuously compounded return is normally distributed. Under this framework, change in asset price is continuous: there are no abrupt jumps.
2. The (continuously compounded) risk-free rate is constant and known. Borrowing and lending are both at the risk-free rate.
3. The volatility of the returns on the underlying asset is constant and known. The price of the underlying changes smoothly (i.e., does not jump abruptly).
4. Markets are “frictionless.” There are no taxes, no transactions costs, and no restrictions on short sales or the use of short-sale proceeds. Continuous trading is possible, and there are no arbitrage opportunities in the marketplace.
5. The (continuously compounded) yield on the underlying asset is constant.

6. The options are European options (i.e., they can only be exercised at expiration).

LOS 29.g: Interpret the components of the Black–Scholes–Merton model as applied to call options in terms of a leveraged position in the underlying.

The formula for valuing a European option using the BSM model is:

$$C_0 = S_0 N(d_1) - e^{-rT} X N(d_2)$$

and

$$P_0 = e^{-rT} X N(-d_2) - S_0 N(-d_1)$$

where:

$$d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)T}{\sigma \sqrt{T}}$$

$$d_2 = d_1 - \sigma \sqrt{T}$$

C_0 and P_0 = values of call and put option

T = time to option expiration

r = continuously compounded risk-free rate

S_0 = current asset price

X = exercise price

σ = annual volatility of asset returns

$N(*)$ = cumulative standard normal probability

$N(-x)$ = $1 - N(x)$

While the BSM model formula looks complicated, its interpretations are not:

1. The BSM value can be thought of as the present value of the expected option payoff at expiration. For a call, that means $C_0 = PV \{S_0 e^{rT} N(d_1) - X N(d_2)\}$. Similarly, for a put option, $P_0 = PV \{X N(-d_2) - S_0 e^{rT} N(-d_1)\}$.
2. Calls can be thought of as a leveraged stock investment where $N(d_1)$ units of stock are purchased using $e^{-rT} X N(d_2)$ of borrowed funds. (A short position in bonds can also be interpreted as borrowing funds.) A portfolio that replicates a put option consists of a long position in $N(-d_2)$ bonds and a short position in $N(-d_1)$ stocks.
3. $N(d_2)$ is interpreted as the risk-neutral probability that a call option will expire in the money. Similarly, $N(-d_2)$ or $1 - N(d_2)$ is the risk-neutral probability that a put option will expire in the money.

EXAMPLE: BSM model

Stock of XZ, Inc., is currently trading at \$50. Suppose that the return volatility is 25% and the continuously compounded risk-free rate is 3%. Calls and puts with a strike price of \$45 and expiring in 6 months ($T = 0.5$) are trading at \$7.00 and \$1.00, respectively. If $N(d_1) = 0.779$ and $N(d_2) = 0.723$, calculate the value of the replicating portfolios and any arbitrage profits on both options.

Answer:

The replicating portfolio for the call can be constructed as long 0.779 shares ($0.779 \times \$50 = \38.95), and borrow $45 \times e^{-0.03(0.5)} \times (0.723) = \32.05 .

$$\text{net cost} = \$38.95 - \$32.05 = \$6.90$$

Because the market price of the call is \$7.00, the profitable arbitrage transaction entails writing a call at \$7.00 and buying the replicating portfolio for \$6.90 to yield an arbitrage profit of \$0.10 per call.

For the put option valuation, note that $N(-d_1) = 1 - N(d_1) = 1 - 0.779 = 0.221$ and $N(-d_2) = 1 - 0.723 = 0.277$.

The replicating portfolio for the put option can be constructed as a long bond position of $45 \times e^{-0.03(0.5)} \times (0.277) = \12.28 and a short position in 0.221 shares resulting in short proceeds of $\$50 \times 0.221$ or \$11.05.

$$\text{net cost} = \$12.28 - \$11.05 = \$1.23$$

Because the market price of the put option is \$1.00, arbitrage profits can be earned by selling the replicating portfolio and buying puts, for an arbitrage profit of \$0.23 per put.

LOS 29.h: Describe how the Black–Scholes–Merton model is used to value European options on equities and currencies.

Options on Dividend Paying Stocks

So far we have assumed that the underlying stock does not pay dividends. If it does, we can adjust the model using a lowercase delta (δ) to represent the dividend yield, as follows:

$$C_0 = S_0 e^{-\delta T} N(d_1) - e^{-rT} X N(d_2)$$

$$P_0 = e^{-rT} X N(-d_2) - S_0 e^{-\delta T} N(-d_1)$$

where:

δ = continuously compounded dividend yield

$$d_1 = \frac{\ln(S/X) + (r - \delta + \sigma^2/2)T}{\sigma \sqrt{T}}$$

$$d_2 = d_1 - \sigma \sqrt{T}$$

Note that $S_0 e^{-\delta T}$ is the stock price, reduced by the present value of any dividends expected to be paid during the option's life.

The put-call parity relation must also be modified if the stock pays dividends:

$$P_0 + S_0 e^{-\delta T} = C_0 + e^{-rT} X$$

Options on Currencies

We can also use the Black-Scholes-Merton model to value foreign exchange options. Here, the underlying is the spot exchange rate instead of a stock price.

The value of an option on a currency can be thought of as being made up of two components, a bond component and a foreign exchange component. The value can be calculated as:

$$C_0 = S_0 e^{-r(B)T} N(d_1) - e^{-r(P)T} X N(d_2)$$

and

$$P_0 = e^{-r(P)T} X N(-d_2) - S_0 e^{-r(B)T} N(-d_1)$$

where:

$r(P)$ = continuously compounded price currency interest rate

$r(B)$ = continuously compounded base currency interest rate

For currencies, the carry benefit is not a dividend but rather interest earned on a deposit of the foreign currency. The spot exchange rate, S_0 , is discounted at the base or foreign currency interest rate, and the bond component, $e^{-r(P)T} X$, is the exercise exchange rate discounted at the price (or domestic currency) interest rate.

LOS 29.i: Describe how the Black model is used to value European options on futures.

The Black Model

If we ignore the mark-to-market feature of the futures contract, The Black model can be used to price European options on forwards and futures:

$$C_0 = e^{-R_f \times T} [F_T \times N(d_1) - X \times N(d_2)]$$

$$d_1 = \frac{\ln(F_T/X) + \left(\frac{\sigma^2}{2}\right)T}{\sigma \sqrt{T}}$$

$$d_2 = d_1 - \sigma \sqrt{T}$$

where:

σ = standard deviation of returns on the futures contract

F_T = futures price

Note that the Black model is just the BSM model with substituted $e^{-R_f \times T} F_T$ for S_0 .

Analogous to the interpretations of the BSM model, an option on futures can be thought of as follows:

- The value of a call option on futures is equal to the value of a portfolio with a long futures position (the PV of the futures price multiplied by $N(d_1)$) and a short bond position (the PV of the exercise price multiplied by $N(d_2)$).
- The value of a put option is equal to the value of a portfolio with a long bond and a short futures position.
- The value of a call can also be thought of as the present value of the difference between the futures price (adjusted by $N(d_1)$) and the exercise price (adjusted by $N(d_2)$).

LOS 29.j: Describe how the Black model is used to value European interest rate options and European swaptions.

Interest Rate Options

Interest rate options are options on forward rates (or options on FRAs). A call option on an FRA gains when rates rise, and a put option on an FRA gains when rates fall. Interest rates are fixed in advance (i.e., at the beginning of the loan term) and settled in arrears (i.e., paid at maturity of the loan). While FRAs generally use a 30/360 convention, options on FRAs use an actual/365 convention.

Consider an interest rate call option on an (M×N) FRA expiring in M months that has a strike rate of X. The underlying is an (N–M) month forward rate. A call option on this FRA can be valued as:

$$C_0 = (AP) e^{-r(\text{actual}/365)} \left[\text{FRA}_{(M \times N)} N(d_1) - XN(d_2) \right] \times NP$$

where:

$$AP = \text{accrual period} = \left[\frac{\text{actual}}{365} \right]$$

NP = notional principal on the FRA

Equivalencies in Interest Rate Derivative Contracts

Combinations of interest rate options can be used to replicate other contracts, for example:

1. A long interest rate call and a short interest rate put (with exercise rate = current FRA rate) can be used to replicate a long FRA (i.e., a forward contract to receive a floating rate and pay a fixed rate).
2. Similarly, if the exercise rate = the current FRA rate, a short interest rate call and long interest rate put can be combined to replicate a short FRA position (i.e., a pay-floating, receive-fixed forward contract).
3. A series of interest rate call options with different maturities and the same exercise price can be combined to form an interest rate cap. (Each of the call options in an interest rate cap is known as a caplet.) A floating rate loan can be hedged using a long interest rate cap.
4. Similarly, an interest rate floor is a portfolio of interest rate put options, and each of these puts is known as a floorlet. Floors can be used to hedge a long position in a floating rate bond.
5. If the exercise rate on a cap and floor is same, a long cap and short floor can be used to replicate a payer swap.
6. Similarly, a short cap and long floor can replicate a receiver swap.
7. If the exercise rate on a floor and a cap are set equal to a market swap fixed rate, the value of the cap will be equal to the value of the floor.

Swaptions

A **swaption** is an option that gives the holder the right to enter into an interest rate swap. A **payer swaption** is the right to enter into a specific swap at some date in the future at a

predetermined rate as the fixed-rate payer. As interest rates increase, the right to take the pay-fixed side of a swap (a payer swaption) becomes more valuable. The holder of a payer swaption would exercise it and enter into the swap if the market rate is greater than the exercise rate at expiration.

A **receiver swaption** is the right to enter into a specific swap at some date in the future as the fixed-rate receiver (i.e., the floating-rate payer) at the rate specified in the swaption. As interest rates decrease, the right to enter the receive-fixed side of a swap (a receiver swaption) becomes more valuable. The holder of a receiver swaption would exercise if market rates are less than the exercise rate at expiration.

A swaption is equivalent to an option on a series of cash flows (annuity), one for each settlement date of the underlying swap, equal to the difference between the exercise rate on the swaption and the market swap fixed rate.

If PVA represents the present value of such an annuity, the value of a payer swaption using the Black model can be calculated as:

$$\text{pay} = (\text{AP}) \text{PVA} [\text{SFR } N(d_1) - X N(d_2)] \text{NP}$$

where:

pay = value of the payer swaption

AP = 1/# of settlement periods per year in the underlying swap

SFR = current market swap fixed rate

X = exercise rate specified in the payer swaption

NP = notional principal of the underlying swap

$$d_1 = \frac{\ln(\text{SFR}/X) + (\sigma^2/2)T}{\sigma \sqrt{T}}$$

$$d_2 = d_1 - \sigma \sqrt{T}$$

The value of a payer swaption is essentially the present value of the expected option payoff:

$$\text{pay} = \text{PV} (E(\text{payoff}))$$

Similarly, the value of a receiver swaption (which we'll call "REC") can be calculated as:

$$\text{REC} = (\text{AP}) \text{PVA} [X N(-d_2) - \text{SFR } N(-d_1)] \text{NP}$$

Equivalencies

A receiver swap can be replicated using a long receiver swaption and a short payer swaption with the same exercise rates. Conversely, a payer swap can be replicated using a long payer swaption and short receiver swaption with the same exercise rates. If the exercise rate is set such that the values of the payer and receiver swaptions are equal, then the exercise rate must be equal to the market forward swap fixed rate.

A long callable bond can be replicated using a long option-free bond plus a short receiver swaption.



MODULE QUIZ 29.5, 29.6

1. Compare the call and put prices on a stock that doesn't pay a dividend (NODIV) with comparable call and put prices on another stock (DIV) that is the same in all respects except it pays a dividend. Which of the following statements is *most accurate*? The price of a:

- A. DIV call will be less than the price of NODIV call.
 - B. NODIV call will equal the price of NODIV put.
 - C. NODIV put will be greater than the price of DIV put.
2. Which of the following is *not* an assumption underlying the Black-Scholes-Merton options pricing model?
- A. The underlying asset does not generate cash flows.
 - B. Continuously compounded returns on the underlying are normally distributed.
 - C. The option can only be exercised at maturity.

Use the following information to answer Questions 3 and 4.

Stock ABC trades for \$60 and has 1-year call and put options written on it with an exercise price of \$60. ABC pays no dividends. The annual standard deviation estimate is 10%, and the continuously compounded risk-free rate is 5%. The value of the call is \$4.09.

Chevron, Inc., common stock trades for \$60 and has a 1-year call option written on it with an exercise price of \$60. The annual standard deviation estimate is 10%, the continuous dividend yield is 1.4%, and the continuously compounded risk-free rate is 5%.

3. The value of the put on ABC stock is *closest* to:
- A. \$1.16.
 - B. \$3.28.
 - C. \$4.09.
4. The value of the call on Chevron stock is *closest* to:
- A. \$3.51.
 - B. \$4.16.
 - C. \$5.61.

MODULE 29.7: OPTION GREEKS AND DYNAMIC HEDGING



Video covering
this content is
available online.

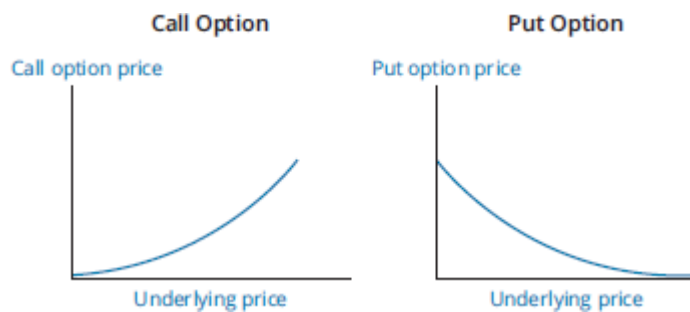
LOS 29.k: Interpret each of the option Greeks.

There are five inputs to the BSM model: asset price, exercise price, asset price volatility, time to expiration, and the risk-free rate. The relationship between each input (except the exercise price) and the option price is captured by sensitivity factors known as “the Greeks.”

Delta describes the relationship between changes in asset prices and changes in option prices. Delta is also the hedge ratio. Call option deltas are positive because as the underlying asset price increases, call option value also increases. Conversely, the delta of a put option is negative because the put value falls as the asset price increases. See Figure 29.4.

Figure 29.4: Call and Put Delta: The Relationship Between Option Price and Underlying

Asset Price



Deltas for call and put options on dividend paying stocks are given by:

$$\text{delta}_C = e^{-\delta T} N(d_1)$$

$$\text{delta}_P = -e^{-\delta T} N(-d_1)$$

where:

δ = continuously compounded dividend yield

However, because we're now using a linear relationship to estimate a non-linear change, the following relationships are only approximations:

$$\Delta C \approx e^{-\delta T} N(d_1) \times \Delta S$$

$$\Delta P \approx -e^{-\delta T} N(-d_1) \times \Delta S$$

where:

ΔC and ΔP = change in call and put price

The approximations are close for small changes in stock price, but the approximation becomes less accurate as the ΔS becomes larger.

EXAMPLE: Calculating change in option price

$e^{-\delta T} N(d_1)$ from the BSM model is 0.58. Calculate the approximate change in the price of a call option on the stock if the stock price increases by \$0.75.

Answer:

$$\Delta C \approx 0.58 \times \$0.75 = \$0.435$$

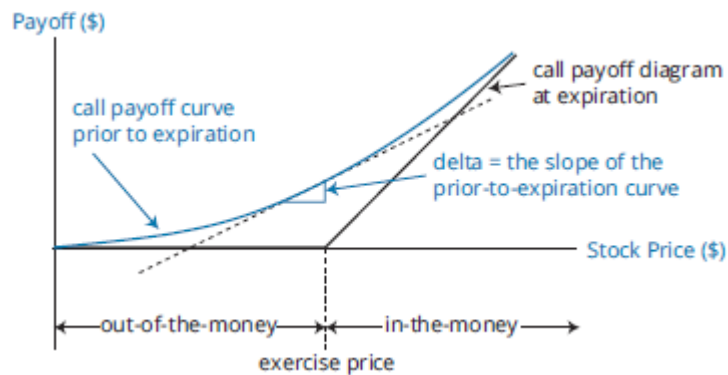
Interpreting Delta

The payoff diagrams for a European call option before- and at-expiration are shown in Figure 29.5. The "at expiration" line represents the call option's intrinsic value, which is equal to:

- Zero when the call option is out-of-the-money.
- The stock price minus the exercise price when the option is in-the-money.

Before expiration, the option also has time value, so the prior-to-expiration curve lies above the at-expiration diagram by the amount of the time value.

Figure 29.5: European Call Option Payoff Diagrams



The slope of the “prior-to-expiration” curve is the change in call price per unit change in stock price. Sound familiar? It should—that’s also the definition of delta. That means *delta is the slope of the prior-to-expiration curve*. The delta of the put option is also the slope of the *prior-to-expiration* put curve.

Take a closer look at Figure 29.5. When the call option is deep out-of-the-money, the slope of the at-expiration curve is close to zero, which means the call delta will be close to zero. For deep out-of-the-money call options (i.e., when the stock price is low), the option price does not change much for a given change in the underlying stock. When the call option is in-the-money (i.e., when the stock price is high), the slope of the at-expiration curve is close to 45 degrees, which means the call delta is close to one. In this case, the call option price will change approximately one-for-one for a given change in the underlying stock price.

The bottom line is that *a call option’s delta will increase from 0 to $e^{-\delta T}$ as stock price increases*. For a non-dividend paying stock, the delta will increase from 0 to 1 as the stock price increases.

For a put option, the put delta is close to zero when the put is out-of-the-money (i.e., when the stock price is high). When the put option is in-the-money (i.e., when the stock price is close to zero), the put delta is close to $-e^{-\delta T}$.

The bottom line is that *a put option’s delta will increase from $-e^{-\delta T}$ to 0 as stock price increases*. For a non-dividend paying stock, the put delta increases from -1 to 0 as the stock price increases.

Now, let’s consider what happens to delta as the option approaches maturity, assuming that the underlying stock price doesn’t change. The effects on call and put options are different and will depend on whether the options are in- or out-of-the-money.

Remember that a call option delta is between 0 and $e^{-\delta T}$. Assuming that the underlying stock price doesn’t change, if the call option is:

- **Out-of-the-money** (the stock price is less than exercise price), the call delta **moves closer to 0** as time passes.
- **In-the-money** (the stock price is greater than exercise price), the call delta **moves closer to $e^{-\delta T}$** as time passes.

Remember that a put option delta is between $-e^{-\delta T}$ and 0. If the put option is:

- **Out-of-the-money** (the stock price is greater than exercise price), the put delta **moves closer to 0** as time passes.
- **In-the-money** (the stock price is less than exercise price), the put delta **moves closer to** $e^{-\delta T}$ as time passes.

Gamma measures the *rate of change in delta* as the underlying stock price changes.

Gamma captures the curvature of the option-value-versus-stock-price relationship. Long positions in calls and puts have positive gammas. For example, a gamma of 0.04 implies that a \$1.00 increase in the price of the underlying stock will cause a call option's *delta* to increase by 0.04, making the call option more sensitive to changes in the stock price.

Gamma is highest for at-the-money options. Deep in-the-money or deep out-of-the-money options have low gamma. Gamma changes with stock price and with time to expiration. To lower (increase) the overall gamma of a portfolio, one should short (go long) options.

Recall that delta provides an approximation for change in option value in response to a change in the price of underlying. Including gamma in our equation would improve the precision with which change in option value would be estimated (similar to how convexity improves the estimate of change in a bond's price in response to a change in interest rate).

$$\Delta C \approx \text{call delta} \times \Delta S + \frac{1}{2} \text{gamma} \times \Delta S^2$$

$$\Delta P \approx \text{put delta} \times \Delta S + \frac{1}{2} \text{gamma} \times \Delta S^2$$

Call and put options on the same underlying asset with the same exercise price and time to expiration will have equal gammas.

Vega measures the sensitivity of the option price to changes in the volatility of returns on the underlying asset, as shown in Figure 29.6. Both call and put options are more valuable, all else equal, the higher the volatility, so vega is positive for both calls and puts. Volatility is a very important input to any option valuation model (and thus vega is an important sensitivity measure) because option values can be very sensitive to changes in volatility. Note that vega gets larger as the option gets closer to being at-the-money.

Figure 29.6: Call and Put Vega: The Relationship Between Option Price and Volatility



Rho measures the sensitivity of the option price to changes in the risk-free rate, as shown in Figure 29.7. The price of a European call or put option does not change much if we use different inputs for the risk-free rate, so rho is not a very important sensitivity measure. Call options increase in value as the risk-free rate increases (the rho of a call option is

positive). You can see this by looking at the BSM model. As the risk-free rate increases, the second term decreases, and the value of the call option increases.

$$C_0 = [S_0 \times N(d_1)] - [X \times e^{-R_f \times T} \times N(d_2)]$$

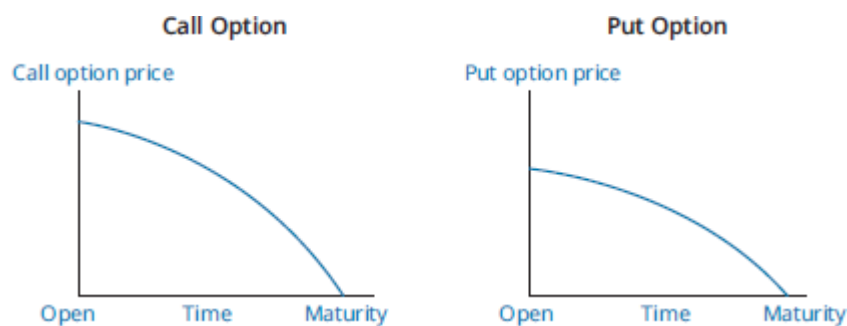
Put options decrease in value as the risk-free rate increases (the rho of a put option is negative).

Figure 29.7: Call and Put Rho: The Relationship Between Option Price and the Risk-Free Rate



Theta measures the sensitivity of option price to the passage of time, as shown in Figure 29.8. As time passes and a call option approaches maturity, its speculative value declines, all else equal. This is called **time decay**. That behavior also applies for *most* put options (though *deep in-the-money* put options close to maturity may actually *increase* in value as time passes). Notice that because it is a measure of time decay, *theta is less than zero*: as time passes and the option approaches the maturity date, the option value decreases (holding other factors constant).

Figure 29.8: Call and Put Theta: The Relationship Between Option Price and the Time to Expiration



PROFESSOR'S NOTE

The statement that theta is less than zero is counterintuitive to many Level II candidates because there are two ways to state the relationship between time and option value:

- The relationship between option value and *time to maturity* is positive: all else equal, shorter maturity options have lower values.
- The relationship between option value and *the passage of time* is negative: all else equal, as time passes and the option approaches maturity, the value of the option decays.

The key to understanding why theta is less than zero is to recognize that it is capturing the intuition of the second statement.

Figure 29.9: Direction of European Option Prices for a Change in the Five Black-Scholes-Merton Model Inputs

Sensitivity Factor (Greek)	Input	Calls	Puts
Delta	Asset price (S)	Positively related Delta > 0	Negatively related Delta < 0
Gamma	Delta	Positive Gamma > 0	Positive Gamma > 0
Vega	Volatility (σ)	Positively related Vega > 0	Positively related Vega > 0
Rho	Risk-free rate (r)	Positively related Rho > 0	Negatively related Rho < 0
Theta	Time to expiration (T)	Time value $\rightarrow$ \$0 as call $\rightarrow$ maturity Theta < 0	Time value $\rightarrow$ \$0 as put $\rightarrow$ maturity Theta < 0*
	Exercise price (X)	Negatively related	Positively related

* There is an exception to the general rule that European put option thetas are negative. The put value may increase as the option approaches maturity if the option is deep in-the-money and close to maturity.

LOS 29.I: Describe how a delta hedge is executed.

Dynamic Hedging

The goal of a **delta-neutral portfolio** (or **delta-neutral hedge**) is to combine a long position in a stock with a *short* position in a call option so that the value of the portfolio does not change as the stock price changes. The number of calls to sell is equal to:

$$\text{number of short call options needed to delta hedge} = \frac{\text{number of shares hedged}}{\text{delta of call option}}$$

Delta-neutral portfolios can also be constructed using put options. In such a case, a long stock position is combined with a *long* position in puts. (Remember that put deltas are negative).

$$\text{number of long put options needed to delta hedge} = - \frac{\text{number of shares}}{\text{delta of the put option}}$$

Delta neutral portfolios are best illustrated with an example.

EXAMPLE: Hedging with call options, part 1

Suppose you own 60,000 shares of Arthurall Company common stock that is currently selling for \$50. A call option on Arthurall with a strike price of \$50 is selling for \$4 and has a delta of 0.60. Determine the number of call options necessary to create a delta-neutral hedge.

Answer:

In order to determine the number of call options necessary to hedge against instantaneous movements in Arthurall's stock price, we calculate:

$$\begin{aligned}\text{number of options needed to delta hedge} &= \frac{60,000}{0.60} \\ &= 100,000 \text{ options, or } 1,000 \text{ option contracts}\end{aligned}$$

Because we are long the stock, we need to short the options.

EXAMPLE: Hedging with call options, part 2

Calculate the effect on portfolio value of a \$1 increase in the price of Arthurall stock.

Answer:

If the price of Arthurall stock increased instantly by \$1.00, the value of your call option position would decrease by \$0.60 because you have sold the call option contracts. Therefore, the net impact of the price change on the value of the hedged portfolio would be zero:

$$\begin{aligned}\text{total change in value of stock position} &= 60,000 \times \$1 = +\$60,000 \\ \text{total change in value of option position} &= 100,000 \times -\$0.60 = -\$60,000 \\ \text{total change in portfolio value} &= \$60,000 - \$60,000 = \$0\end{aligned}$$

A key consideration in delta-neutral hedging is that the delta hedged asset position is only risk free for a very small change in the value of the underlying stock. The delta-neutral portfolio must be continually rebalanced to maintain the hedge; for this reason, it is called a dynamic hedge. As the underlying stock price changes, so does the delta of the call option, and thus too the number of calls that need to be sold to maintain a hedged position. Hence, continuously maintaining a delta-neutral position involves significant transaction costs.

LOS 29.m: Describe the role of gamma risk in options trading.

If the assumptions of the BSM hold, changes in stock price will be continuous rather than abrupt, and hence there will be no gamma risk. In this context, gamma can be viewed as a measure of how poorly a dynamic hedge will perform when it is not rebalanced in

response to a change in the asset price. **Gamma risk** is therefore the risk that the stock price might abruptly “jump,” leaving an otherwise delta-hedged portfolio unhedged.

Consider a delta hedge involving a long position in stock and short position in calls. If the stock price falls abruptly, the loss in the long stock position will not equal the gain in the short call position. This is the gamma risk of the hedge.

Because a stock’s delta is always 1, its gamma is 0. A delta-hedged portfolio with a long position in stocks and a short position in calls will have negative net gamma exposure.

LOS 29.n: Define implied volatility and explain how it is used in options trading.

Future volatility is one of the inputs in the BSM model. Estimates based on historical volatility are often used because future volatility is not directly observable.

Implied volatility is the standard deviation of continuously compounded asset returns that is “implied” by the market price of the option. Of the five inputs into the BSM model, four are observable: stock price, exercise price, risk-free rate, and time to maturity. If we plug these four inputs into the BSM model, we can solve for the value of volatility that makes the BSM model value equal the market price.

Volatility enters the BSM pricing equation in a complex way, and there is no closed-form solution for the volatility that will produce a value equal to the market price. Rather, implied volatility must be found by iteration (i.e., trial and error).

Traders often use implied volatilities to gauge market perceptions. For example, implied volatilities in options with different exercise prices on the same underlying may reflect different implied volatilities (a violation of the BSM assumption of constant volatility).

Traders can also use implied volatility as a mechanism to quote option prices. In this way, options with different exercise prices and maturity dates can be quoted using the same unit of measurement.

EXAMPLE: Implied volatility

Calls on Blue, Inc., stock are currently trading at an implied volatility of 22%. A trader estimates that the future volatility will actually be closer to 25%. To capitalize on her beliefs, the trader should:

- A. change her model.
- B. buy calls on Blue stock.
- C. write calls on Blue stock.

Answer:

- B. Based on the trader’s beliefs, call options on Blue stock are underpriced in the market. Accordingly, she should buy the calls.



MODULE QUIZ 29.7

1. Which of the following statements is *least accurate*? The value of a:

- A. call option will decrease as the risk-free rate increases.
 B. put option will decrease as the exercise price decreases.
 C. call option will decrease as the underlying stock price decreases.
2. Which of the following inputs into the Black-Scholes-Merton model is *least likely* to have opposite effects on put and call prices?
 A. Volatility.
 B. Strike price.
 C. Risk-free rate.
3. Which of the following statements is *most accurate*? Implied volatility:
 A. requires market prices.
 B. requires a series of past returns.
 C. is equal for otherwise identical options with different maturities.
4. The delta of a put is -0.43 . If the price of the underlying stock increases from \$40 to \$44, the price of the put option:
 A. increases by approximately 4.3%.
 B. decreases by approximately 4.3%.
 C. decreases by approximately \$1.72.
5. The current stock price of Heart, Inc., is \$80. Call and put options with exercise prices of \$50 and 15 days to maturity are currently trading. Which of these scenarios is *most likely* to occur if the stock price falls by \$1?
- | <u>Call value</u> | <u>Put value</u> |
|-----------------------|--------------------|
| A. Decrease by \$0.94 | Increase by \$0.08 |
| B. Decrease by \$0.76 | Increase by \$0.96 |
| C. Decrease by \$0.07 | Increase by \$0.89 |
6. A put option with an exercise price of \$45 is trading for \$3.50. The current stock price is \$45. What is the *most likely* effect on the option's delta and gamma if the stock price increases to \$50?
 A. Both delta and gamma will increase.
 B. Both delta and gamma will decrease.
 C. One will increase and the other will decrease.
7. From the Black-Scholes-Merton model, $N(d_1) = 0.42$ for a 3-month call option on Panorama Electronics common stock. If the stock price falls by \$1.00, the price of the call option will:
 A. decrease by less than the increase in the price of the put option.
 B. increase by more than the decrease in the price of the put option.
 C. decrease by the same amount as the increase in the price of the put option.

KEY CONCEPTS

LOS 29.a

To value an option using a two-period binomial model:

- Calculate the stock values at the end of two periods (there are three possible outcomes because an up-down move gets you to the same place as a down-up move).
- Calculate option payoffs at the end of two periods.
- Calculate expected values at the end of two periods using the up- and down-move probabilities. Discount these back one period at the risk-free rate to find the option values at the end of the first period.

- Calculate expected value at the end of period one using the up- and down-move probabilities. Discount this back one period to find the option value today.

To price an option on a bond using a binomial tree, (1) price the bond at each node using projected interest rates, (2) calculate the intrinsic value of the option at each node at maturity of the option, and (3) calculate the value of the option today.

LOS 29.b

Option values can be calculated as present value of expected payoffs on the option, discounted at the risk-free rate. The probabilities used to calculate the expected value are risk-neutral probabilities.

LOS 29.c

Synthetic call and put options can be created using a replicating portfolio. A replication portfolio for a call option consists of a leveraged position in h shares where h is the hedge ratio or delta of the option. A replication portfolio for a put option consists of a long position in a risk-free bond and a short position in h shares. If the value of the option exceeds the value of the replicating portfolio, an arbitrage profit can be earned by writing the option and purchasing the replicating portfolio.

LOS 29.d

The value of a European call option using the binomial option valuation model is the present value of the expected value of the option in the 2 states.

$$C_0 = \frac{(\pi_U \times C^+ + \pi_D \times C^-)}{(1 + R_f)}$$

The value of an American-style call option on a non-dividend paying stock is the same as the value of an equivalent European-style call option. American-style put options may be more valuable than equivalent European-style put options due to the ability to exercise early and earn interest on the intrinsic value.

LOS 29.e

The value of an interest rate option is computed similarly to the value of options on stocks: as the present value of the expected future payoff. Unlike binomial stock price trees, binomial interest rate trees have equal (risk-neutral) probabilities of the up and down states occurring.

LOS 29.f

The assumptions underlying the BSM model are:

- The price of the underlying asset changes smoothly (i.e., does not jump) and has a normally distributed continuously compounded return.
- The (continuous) risk-free rate is constant and known.
- The volatility of the underlying asset is constant and known.
- Markets are “frictionless.”
- The continuously compounded yield on the underlying asset is constant.
- The options are European.

LOS 29.g

Calls can be thought of as leveraged stock investment where $N(d_1)$ units of stock is purchased using $e^{-rT}XN(d_2)$ of borrowed funds. A portfolio that replicates a put option can be constructed by combining a long position in $N(-d_2)$ bonds and a short position in $N(-d_1)$ stocks.

LOS 29.h

European options on dividend-paying stock can be valued by adjusting the model to incorporate the yield on the stock: the current stock price is adjusted by subtracting the present value of dividends expected up until option expiration. Options on currencies incorporate a yield on the foreign currency based on the interest rate in that currency.

LOS 29.i

The Black model is simply the BSM model with $e^{-R_f \times T} F_T$ substituted for S_0 .

LOS 29.j

The Black model can be used to value interest rate options by substituting the current forward rate in place of the stock price and the exercise rate in place of exercise price. The value is adjusted for accrual period (i.e., the period covered by the underlying rate).

A swaption is an option that gives the holder the right to enter into an interest rate swap. A payer (receiver) swaption is the right to enter into a swap as the fixed-rate payer (receiver). A payer (receiver) swaption gains value when interest rates increase (decrease).

LOS 29.k

Direction of BSM option value changes for an increase in the five model inputs:

Sensitivity Factor (Greek)	Input	Calls	Puts
Delta	Asset price (S)	Positively related Delta > 0	Negatively related Delta < 0
Gamma	Delta	Positively related Gamma > 0	Positively related Gamma > 0
Vega	Volatility (σ)	Positively related Vega > 0	Positively related Vega > 0
Rho	Risk-free rate (r)	Positively related Rho > 0	Negatively related Rho < 0
Theta	Time to expiration (T)	Time value $\rightarrow$ \$0 as call $\rightarrow$ maturity Theta < 0	Time value $\rightarrow$ \$0 as put $\rightarrow$ maturity Theta < 0
	Exercise price (X)	Negatively related	Positively related

Delta is the change in the price of an option for a one-unit change in the price of the underlying security. $e^{-\delta T}N(d_1)$ from the BSM model is the delta of a call option, while $-e^{-\delta T}N(-d_1)$ is the put option delta.

As stock price increases, delta for a call option increases from 0 to $e^{-\delta T}$, while delta for a put option increases from $-e^{-\delta T}$ to 0.

LOS 29.1

The goal of a delta-neutral portfolio (or delta-neutral hedge) is to combine a long position in a stock with a short position in call options (or a long position in put options) so that the portfolio value does not change when the stock value changes. Given that delta changes when stock price changes, a delta hedged portfolio needs to be continuously rebalanced. Gamma measures how much delta changes as the asset price changes and, thus, offers a measure of how poorly a fixed hedge will perform as the price of the underlying asset changes.

LOS 29.m

When the price of the underlying stock abruptly jumps, a violation of BSM, the delta of the option would change (captured by the option gamma), leaving a previously delta hedged portfolio unhedged. This is the gamma risk of a delta hedged portfolio.

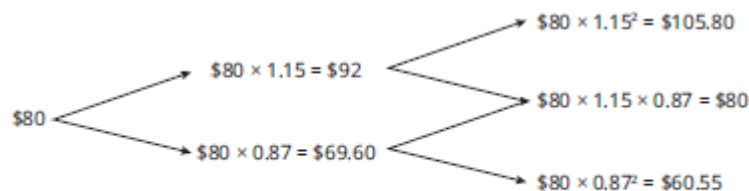
LOS 29.n

Implied volatility is the volatility that, when used in the Black-Scholes formula, produces the current market price of the option. If an option is overvalued, implied volatility is too high.

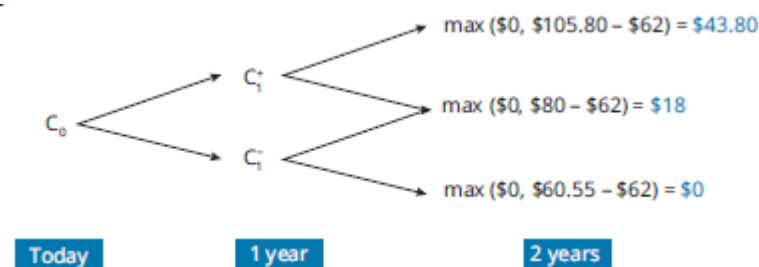
ANSWER KEY FOR MODULE QUIZZES

Module Quiz 29.1, 29.2

1. B Stock Tree



Option Tree



$$U = 1.15$$

$$D = 0.87$$

$$\pi_U = \frac{1.04 - 0.87}{1.15 - 0.87} = 0.61$$

$$\pi_D = 1 - 0.61 = 0.39$$

$$C_2^{++} = \$43.80$$

$$C_2^{+-} = C_2^{-+} = \$18.00$$

$$C_2^{--} = \$0$$

$$C_1^{+} = \frac{(0.61 \times \$43.80) + (0.39 \times \$18.00)}{1.04} = \$32.44$$

$$C_1^{-} = \frac{(0.61 \times \$18.00) + (0.39 \times \$0)}{1.04} = \$10.56$$

$$C_0 = \frac{(0.61 \times \$32.44) + (0.39 \times \$10.56)}{1.04} = \$22.99$$

(Module 29.2, LOS 29.d)

Module Quiz 29.3

1. **B** In the upper node at the end of the first year, the European option is worth \$0.29, but the American option can be exercised and a profit of \$0.57 realized (the difference between the bond price of \$100.57 and the exercise price of \$100).

In the lower node at the end of the first year, the European option is worth \$1.35, but the American option can be exercised and a profit of \$3.80 realized (the difference between the bond price of \$103.80 and the exercise price of \$100).

The value of the American option today is therefore:

$$\text{American option price} = \frac{(\$0.57 \times 0.5) + (\$3.80 \times 0.5)}{1.03} = \$2.12$$

(LOS 29.d)

Module Quiz 29.4

1. **C** The hedge ratio in a one-period model is equivalent to a delta, the ratio of the call price change to the stock price change. We will sell the 1,000 calls because they are overpriced. Buying 350 shares of stock will produce a riskless hedge. The payoff at expiration will return more than the riskless rate on the net cost of the hedge portfolio. Borrowing to finance the hedge portfolio and earning a higher rate than the borrowing rate produces arbitrage profits. (LOS 29.c)

2. **A** A synthetic European put option is formed by:

- Buying a European call option.
- Short-selling the stock.
- Buying (i.e., investing) the present value of the exercise price worth of a pure-discount riskless bond.

(LOS 29.c)

Module Quiz 29.5, 29.6

1. **A** The dividend affects option values because if you own the option, you do not have access to the dividend. Hence, if the firm pays a dividend during the life of the option, this must be considered in the valuation formula. Dividends decrease the value of call options, all else equal, and they increase the value of put options. (Module 29.6, LOS 29.h)
2. **A** To derive the BSM model, we need to assume no arbitrage is possible and that:
 - A continuously compounded return on the underlying has a normal distribution.
 - The (continuous) risk-free rate is constant.
 - The volatility of the underlying asset is constant.
 - Markets are “frictionless.”
 - The (continuously compounded) yield on the underlying asset is constant (and not necessarily zero).
 - The options are European (i.e., they can only be exercised at maturity).(Module 29.6, LOS 29.f)

3. **A** According to put/call parity, the put's value is:

$$P_0 = C_0 - S_0 + (X \times e^{-R_f \times T}) = \$4.09 - \$60.00 + [\$60.00 \times e^{-(0.05 \times 1.0)}] = \$1.16$$

(Module 29.6, LOS 29.h)

4. **A** ABC and Chevron stock are identical in all respects except Chevron pays a dividend. Therefore, the call option on Chevron stock must be worth less than the call on ABC (i.e., less than \$4.09). \$3.51 is the only possible answer. (Module 29.6, LOS 29.h)

Module Quiz 29.7

1. **A** The value of a call and the risk-free rate are positively related, so as the risk-free rate increases, the value of the call will increase. (LOS 29.k)
2. **A** Volatility increases will increase the values of both puts and calls. (LOS 29.k)
3. **A** Implied volatility is the volatility that produces market option prices from the BSM model. Its use for pricing options is limited because it is based on market prices. Past returns are used to calculate historical volatility. (LOS 29.n)
4. **C** The put option will decrease in value as the underlying stock price increases: $-0.43 \times \$4 = -\1.72 . (LOS 29.k)
5. **A** The call option is deep in-the-money and must have a delta close to one. The put option is deep out-of-the-money and will have a delta close to zero. Therefore, the value of the in-the-money call will decrease by close to \$1 (e.g., \$0.94), and the value of the out-of-the-money put will increase by a much smaller amount (e.g., \$0.08). The call price will fall by more than the put price will increase. (LOS 29.k)
6. **C** The put option is currently at-the-money because its exercise price is equal to the stock price of \$45. As stock price increases, the put option's delta (which is less than zero) will increase toward zero, becoming less negative. The put option's gamma,

which measures the rate of change in delta as the stock price changes, is at a maximum when the option is at-the-money. Therefore, as the option moves out-of-the-money, its gamma will fall. (LOS 29.k)

7. **A** If $\Delta S = -\$1.00$, $\Delta C \approx 0.42 \times (-1.00) = -\0.42 , and $\Delta P \approx (0.42 - 1) \times (-1.00) = \0.58 , the call will decrease by less (\$0.42) than the increase in the price of the put (\$0.58). (LOS 29.k)

Topic Quiz: Derivatives

You have now finished the Derivatives topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow three minutes per question.